

CSEE 3827: Fundamentals of Computer Systems, Spring 2026

Lecture 3

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Agenda (M&K 2.3-2.5)

- Standard Forms
 - Product-of-Sums (PoS)
 - Sum-of-Products (SoP)
 - converting between
 - Min-terms and Max-terms
- Simplification via Karnaugh Maps (K-maps)
 - 2, 3, and 4 variable
 - Implicants, Prime Implicants, Essential Prime Implicants
 - Using K-maps to reduce
 - PoS form
 - Don't Care Conditions

Standard Forms

- There are many ways to express a boolean expression

$$\begin{aligned} F &= XYZ + XYZ + XZ \\ &= XY(Z + Z) + XZ \\ &= XZ \end{aligned}$$

- Generally: want most simplified (fewest # of literals)
- Sometimes, useful to consider other forms:
 - Can help lead to a simplified form
 - Can help with understanding of a particular function

Two principal standard forms

- Sum-of-products (SoP)
- Product-of-sums (PoS)
- We will deal mostly in this course with SoP

Product and sum terms

- Product term: logical AND of literals (e.g., $X\bar{Y}Z$)
- Sum term: logical OR of literals (e.g., $A + \bar{B} + C$)

Understanding Product and Sum Terms

- A **product term** evaluates to true only when **all literals** in the product term are True
 - X = study hard
 - Y = show up late for interview
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 - (study hard) AND (don't show up late to interview) AND be a team player
 - i.e., $(X=1, Y=0, Z=1)$
 - i.e., X is True and Y is false and Z is true

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 - A = Ace Exams
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 - C = Columbia removes "F" and "D" designations

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 - A = Ace Exams
 - B = skip >20% of P-credit OH
 - C = Columbia removes "F" and "D" designations —
 - To pass fundies: $A + B + C$
 - i.e., (Ace Exams) OR (don't skip 20% of P-credit OH) OR (Columbia removes D,F)
 - i.e., A is True or B is false or C is true

Variables that “don’t matter”

- There may be other variables in the equation. If their value doesn’t matter, we can leave them out
 - X = study hard (SH)
 - Y = show up late for interview (SL)
 - Z = be a team player (TP)
 - W = wear a suit and tie (ST)

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 - i.e., [SH & don’t SL & TP & ST] OR [SH & don’t SL & TP & don’t ST]
 - Recall:
 - $WX\bar{Y}Z + \bar{W}X\bar{Y}Z = (W + \bar{W}) X\bar{Y}Z = X\bar{Y}Z$

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 - $WX\bar{Y}Z + \bar{W}X\bar{Y}Z = (W + \bar{W})X\bar{Y}Z = X\bar{Y}Z$

W	X	Y	Z	S
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

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0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Variables that “don’t matter”

- There may be other variables in the equation. If their value doesn’t matter, we can leave them out

- X = study hard (SH)
- Y = show up late for interview (SL)
- Z = be a team player (TP)
- W = wear a suit and tie (ST)

- Secret to success: $WX\bar{Y}Z + \bar{W}X\bar{Y}Z$
- i.e., [SH & don’t SL & TP & ST] OR [SH & don’t SL & TP & don’t ST]
- Recall:
 - $WX\bar{Y}Z + \bar{W}X\bar{Y}Z = (W + \bar{W}) X\bar{Y}Z = X\bar{Y}Z$

W	X	Y	Z	S
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Wear a suit & study hard & show up late & don’t be team player → not successful!

Variables that “don’t matter”

- There may be other variables in the equation. If their value doesn’t matter, we can leave them out

- X = study hard (SH)
- Y = show up late for interview (SL)
- Z = be a team player (TP)
- W = wear a suit and tie (ST)

- Secret to success: $WX\bar{Y}Z + \bar{W}X\bar{Y}Z$
- i.e., [SH & don’t SL & TP & ST] OR [SH & don’t SL & TP & don’t ST]
- Recall:
 - $WX\bar{Y}Z + \bar{W}X\bar{Y}Z = (W + \bar{W}) X\bar{Y}Z = X\bar{Y}Z$

Values in blue are what “force” S = 0. Only the rows with no blues result in S=1

W	X	Y	Z	S
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Variables that “don’t matter”: Example 2

- There may be other variables in the equation. If their value doesn’t matter, we can leave them out

- A = Ace Exams
- B = skip >20% of P-credit OH
- C = Columbia removes “F” and “D” designations
- D = Skip lunch today
- To pass fundies: $A + \bar{B} + C$

A	B	C	D	P
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Inputs shown in red are what “force” P to be True

PoS & SoP

- Sum of products (SoP): OR of ANDs (OR of product terms)

$$\text{e.g., } F = \bar{Y} + \bar{X}Y\bar{Z} + XY$$

Order of precedence in this form: do all the ANDs before any ORs (don't need parentheses)

- Product of sums (PoS): AND of ORs (AND of sum terms)

$$\text{e.g., } G = X(\bar{Y} + Z)(X + Y + \bar{Z})$$

Order of precedence in this form: do all the ORs before any ANDs (Need parentheses)

Both forms at same time!

- If only have ANDs or only have ORs, then can be in both forms at once
- e.g., $A + B + \overline{C}$ is both SoP and PoS (3 SoP terms, 1 literal each; 1 PoS term, 3 literals)
- e.g., $AB\overline{C}$ also both SoP and PoS (3 PoS terms, 1 literal each; 1 SoP term, 3 literals)

PoS and SoP not always simplest form

- e.g., $F = ABD + ABE + C(D+E)$
 - $(AB+C)(D+E)$ is **simplest** (fewest literals) form (5 literals)
 - know it's simplest because each literal appears only once
 - but not in SoP form or in PoS form
 - Why not? We AND first (AB) then OR (with C), then AND again (with D+E)
 - Turns out:
 - simplest SoP form: $ABD + ABE + CD + CE$ (10 literals)
 - simplest PoS form: $(A+C)(B+C)(D+E)$ is (6 literals)
 - Looking Ahead: SoP form will be important for this course!

Converting any expression to SoP

- Just “multiply” through and simplify

- e.g., $G = X(Y + Z)(X + Y + Z)$

$$= XYX + XYY + XYZ + XZX + XZY + XZZ \quad \text{and then simplify}$$

$$= XY + XY + XYZ + XZ + XZY + XZ \quad (\text{removed duplicate literals in prod. term})$$

$$= XY + XZ \quad (\text{removed repeated prod. terms})$$

This will get expression in SoP form, but it might not be the simplest SoP form

$$\text{e.g., } (X+Y)(X+Z) = XX + XZ + YX + YZ = X + XZ + YX + YZ$$

This can in fact be simplified to $X + YZ$ (remember $X+XZ = X$, $X+YX=X$)

Converting from SoP to PoS

- Complement, multiply through, complement (swap + and $\cdot$ ops, flip literals)
- e.g., $F = \bar{Y}\bar{Z} + X\bar{Y}Z + XY\bar{Z}$
 - $\bar{F} = (Y+Z)(\bar{X} + Y + \bar{Z})(\bar{X} + \bar{Y} + Z)$ swap ops, flip literals
 - $= YZ + \bar{X}Y + \bar{X}\bar{Z}$ (after lots of simplifying) simplify $\bar{F}$
 - $F = (\bar{Y}+\bar{Z})(X+\bar{Y})(X+\bar{Z})$ swap ops, flip literals
- Why did this work?
 - Since $\bar{F} = YZ + \bar{X}Y + \bar{X}\bar{Z}$ in SoP form, by DeMorgan, or Dual + flip literals:
 - Complementing $\bar{F}$ in SoP form yields F in PoS form

Minterms and Maxterms

Minterms

e.g., Minterms for 3 variables A,B,C

A	B	C	minterm
0	0	0	m0 $\bar{A}\bar{B}\bar{C}$
0	0	1	m1 $\bar{A}\bar{B}C$
0	1	0	m2 $\bar{A}B\bar{C}$
0	1	1	m3 $\bar{A}BC$
1	0	0	m4 $A\bar{B}\bar{C}$
1	0	1	m5 $A\bar{B}C$
1	1	0	m6 $AB\bar{C}$
1	1	1	m7 ABC

- A **product term** in which **all variables appear exactly once**, either complemented or uncomplemented.
- Each minterm evaluates to 1 for **exactly one** assignment of values to all variables (one row in truth table), 0 for all others.
- e.g., for what values of A,B,C does $\bar{A}\bar{B}C$ evaluate to 1?

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- A **product term** in which **all variables appear exactly once**, either complemented or uncomplemented.
- Each minterm evaluates to 1 for **exactly one** assignment of values to all variables (one row in truth table), 0 for all others.
- e.g., for what values of A,B,C does $\bar{A}\bar{B}C$ evaluate to 1?
- Ans: A=0, B=0, C=1 (that's the only one)
E.g., A=0, B=0, C=0, $\bar{A}\bar{B}C = 0$ (because of C)

Minterms

e.g., Minterms for 3 variables A,B,C

A	B	C	minterm
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0	0	1	m1 $\bar{A}\bar{B}C$
0	1	0	m2 $\bar{A}B\bar{C}$
0	1	1	m3 $\bar{A}BC$
1	0	0	m4 $A\bar{B}\bar{C}$
1	0	1	m5 $A\bar{B}C$
1	1	0	m6 $AB\bar{C}$
1	1	1	m7 ABC

- A **product term** in which **all variables appear exactly once**, either complemented or uncomplemented.
- Each minterm evaluates to 1 for **exactly one** assignment of values to all variables (one row in truth table), 0 for all others.
- Each product term denoted by mX where X corresponds to the row of the truth table where variable value assignments cause that minterm to equal 1.

Minterm examples (with 3 variables, A,B,C)

- ABC is a minterm, and is true when and only when $A=1, B=1, C=1$
- $\overline{A}BC$ is a minterm and is true when and only when $A=0, B=1, C=0$
- A function can be described as a **sum of its minterms**
 - e.g., $F = B(A \oplus \overline{C}) = B(AC + \overline{A}\overline{C}) = ABC + \overline{A}B\overline{C}$
 - $F=1$ when ABC is true OR $\overline{A}B\overline{C}$ is true
 - i.e., $F = 1$ when $A=1 \ \& \ B=1 \ \& \ C=1$ OR when $A=0 \ \& \ B=1 \ \& \ C=0$
 - $F = 0$ otherwise

F evaluates to 1 for a given assignment of inputs if and only if its sum of minterms contains the minterm for that assignment

A few more Minterm examples (of 3 variables A,B,C)

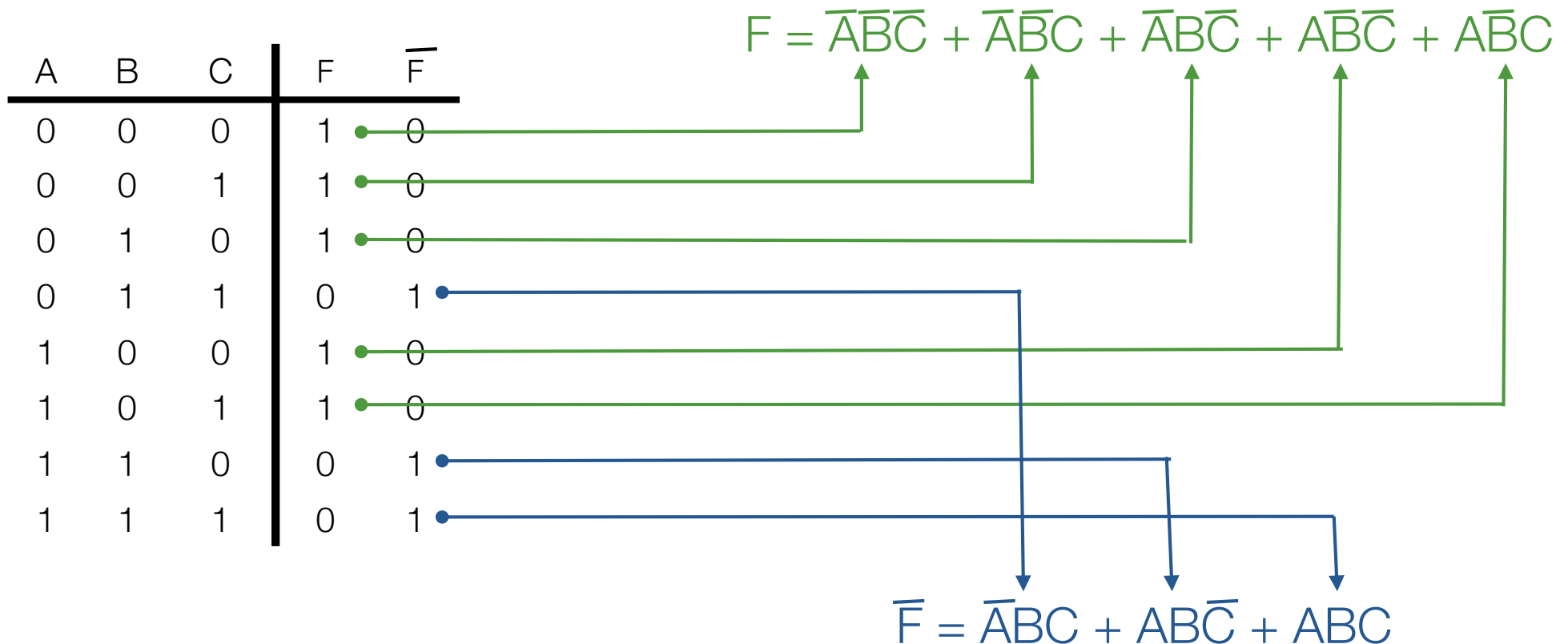
- $G(A,B,C) = BC$ (G's value is independent of A)
 - $G = ABC + \overline{A}BC$ (G is defined over A,B,C, minterms contain all vars)
 - $G = 1$ when & only when $A=1,B=1,C=1$ OR $A=0,B=1,C=1$
- $H(A,B,C) = A$
 - $H = ABC + A\overline{B}\overline{C} + A\overline{B}C + A\overline{B}\overline{C}$ (all combos of B,C when $A=1$)
- $F = A + BC$
 - $= H + G = \underbrace{\overline{A}BC + \overline{A}\overline{B}C + \overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C}}_A + \underbrace{ABC + \overline{A}BC}_{BC}$ (note minterm ABC repeated)

Minterms to describe a function

- sometimes also called a **minterm expansion: OR (SUM) appropriate minterms together**

This “term” is TRUE when A=0,B=1,C=0

Hence F=1 when A=0,B=1,C=0



Function and its complement function use all minterms, share none in common

Sum of minterms form

- The logical OR of all minterms for which $F = 1$.

A	B	C	minterm	F
0	0	0	m0 $\bar{A}\bar{B}\bar{C}$	0
0	0	1	m1 $\bar{A}\bar{B}C$	1
0	1	0	m2 $\bar{A}B\bar{C}$	1
0	1	1	m3 $\bar{A}BC$	1
1	0	0	m4 $A\bar{B}\bar{C}$	0
1	0	1	m5 $A\bar{B}C$	0
1	1	0	m6 $AB\bar{C}$	0
1	1	1	m7 ABC	0

$$\begin{aligned}F &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC \\&= m1 + m2 + m3 \\&= \sum m(1,2,3)\end{aligned}$$

Minterm form cont'd

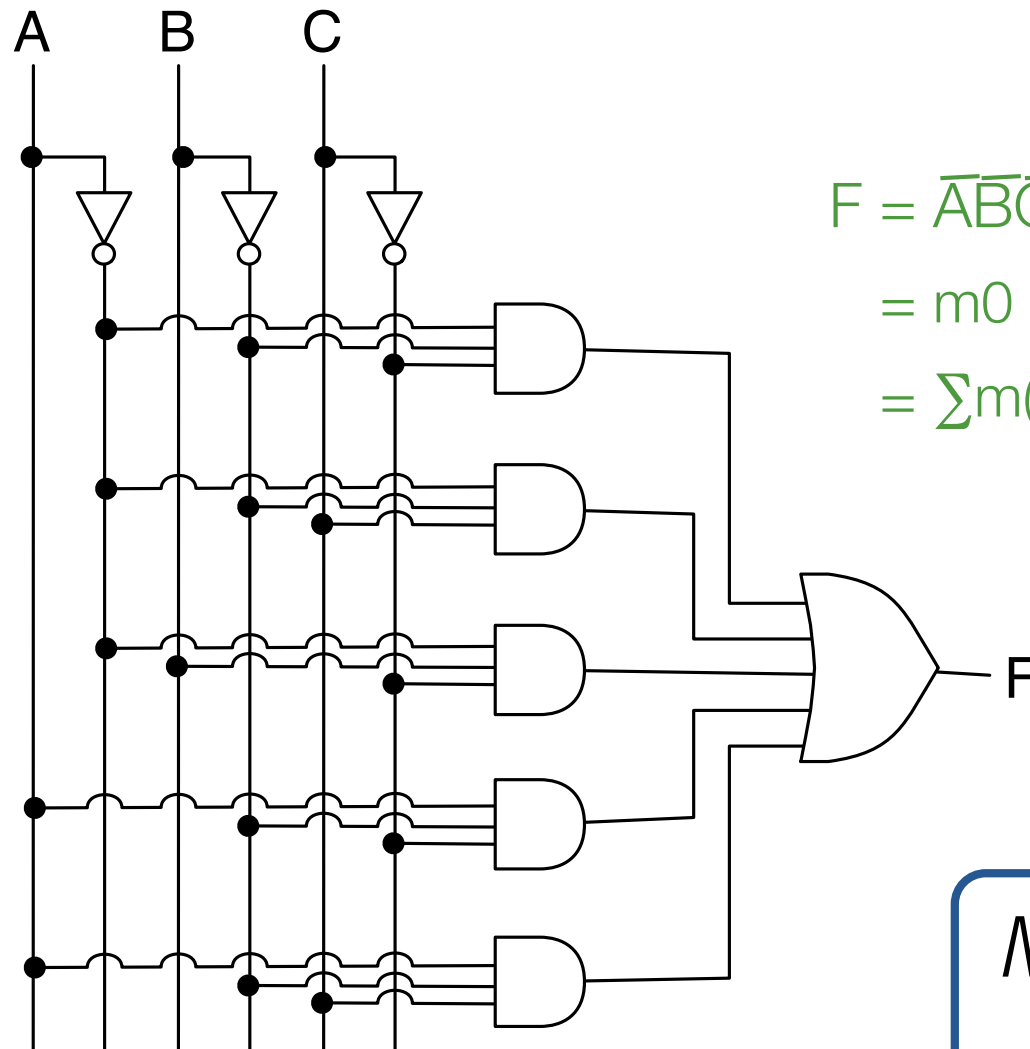
(variables appear once in each minterm)

A	B	C	F	$\overline{F}$	minterm
0	0	0	1	0	m0 $\overline{A}\overline{B}\overline{C}$
0	0	1	1	0	m1 $\overline{A}\overline{B}C$
0	1	0	1	0	m2 $\overline{A}B\overline{C}$
0	1	1	0	1	m3 $\overline{A}BC$
1	0	0	1	0	m4 $A\overline{B}\overline{C}$
1	0	1	1	0	m5 $A\overline{B}C$
1	1	0	0	1	m6 $AB\overline{C}$
1	1	1	0	1	m7 ABC

$$\begin{aligned}
 F &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + A\overline{B}C \\
 &= m0 + m1 + m2 + m4 + m5 \\
 &= \sum m(0,1,2,4,5)
 \end{aligned}$$

$$\begin{aligned}
 \overline{F} &= \overline{A}BC + AB\overline{C} + ABC \\
 &= m3 + m6 + m7 \\
 &= \sum m(3,6,7)
 \end{aligned}$$

Minterms as a circuit



$$\begin{aligned} F &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C \\ &= m_0 + m_1 + m_2 + m_4 + m_5 \\ &= \sum m(0,1,2,4,5) \end{aligned}$$

*Minterm form is not
simplified form!*

Simplest Form v. SoP Form v. Minterm Form

- Can be the same, but not always
- e.g., $F = WX(Y+Z)$ (note 4 variables)
 - SoP form: $WXY + WXZ$
 - Minterm form: $WXYZ + WXY\bar{Z} + WX\bar{Y}Z$
- e.g., $F = WX(Y\bar{Z} + \bar{Y}Z)$
 - SoP form: $WX\bar{Y}Z + WXY\bar{Z}$
 - Minterm form: $WX\bar{Y}Z + WXY\bar{Z}$

Maxterms - “Dual” of minterms

A	B	C	maxterm
0	0	0	M0 $A+B+C$
0	0	1	M1 $A+B+\bar{C}$
0	1	0	M2 $A+\bar{B}+C$
0	1	1	M3 $A+\bar{B}+\bar{C}$
1	0	0	M4 $\bar{A}+B+C$
1	0	1	M5 $\bar{A}+B+\bar{C}$
1	1	0	M6 $\bar{A}+\bar{B}+C$
1	1	1	M7 $\bar{A}+\bar{B}+\bar{C}$

- A sum term in which all variables appear once, either complemented or uncomplemented.
- Each maxterm evaluates to 0 for exactly one variable assignment, 1 for all others.
- Denoted by MX where X corresponds to the variable assignment for which $MX = 0$.

Maxterms: not as intuitive as minterms

- $F = \overline{A}\overline{C} + \overline{B}$
 - turns out $F = 1$ when $(A=1 \text{ or } B=0 \text{ or } C=0)$ AND $(A=0 \text{ or } B=0 \text{ or } C=1)$ AND $(A=0 \text{ or } B=0 \text{ or } C=0)$,
 - i.e., $F = (A+\overline{B}+\overline{C})(\overline{A}+\overline{B}+C)(\overline{A}+\overline{B}+\overline{C})$
- Think of it as forcing F to 0 when F doesn't equal a maxterm
 - e.g., $F = 0$ when $A=1, B=1, C=1$.
 - In other words, for $F=1$, it is necessary (but not sufficient) that either $A=0$ OR $B=0$ OR $C=0$, i.e., $(\overline{A}+\overline{B}+\overline{C} = 1)$ AND some other conditions
 - Thus, the maxterm $(\overline{A}+\overline{B}+\overline{C})$ is included in the product for F

Maxterm description of a function

- sometimes also called **conjunctive normal form (CNF)**
- sometimes also called a **maxterm expansion**

This “term” is FALSE when
 $A=1 \ \& \ B=1 \ \& \ C=0$

A	B	C	F	$\overline{F}$
0	0	0	1	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	0	1
1	1	1	0	1

$$F = (A + \overline{B} + \overline{C}) (\overline{A} + \overline{B} + C) (\overline{A} + \overline{B} + \overline{C})$$

Force to 0

When inputs “satisfy” a Maxterm, the function equals 0

Product of maxterms form

- The logical AND of all maxterms for which $F = 0$.

A	B	C	maxterm	F
0	0	0	M0 $A+B+C$	0
0	0	1	M1 $A+B+\bar{C}$	1
0	1	0	M2 $A+\bar{B}+C$	1
0	1	1	M3 $A+\bar{B}+\bar{C}$	1
1	0	0	M4 $\bar{A}+B+C$	0
1	0	1	M5 $\bar{A}+B+\bar{C}$	0
1	1	0	M6 $\bar{A}+\bar{B}+C$	0
1	1	1	M7 $\bar{A}+\bar{B}+\bar{C}$	0

$$\begin{aligned} F &= (A+B+C) (\bar{A}+B+C) (\bar{A}+B+\bar{C}) (\bar{A}+\bar{B}+C) (\bar{A}+\bar{B}+\bar{C}) \\ &= (M0) (M4) (M5) (M6) (M7) \\ &= \prod M(0,4,5,6,7) \end{aligned}$$

Summary of Minterms and Maxterms

	F	$\overline{F}$
Minterms (SOP)	$\sum m(F = 1)$	$\sum m(F = 0)$
Maxterms (POS)	$\prod M(F = 0)$	$\prod M(F = 1)$

One final example

A	B	C	F	$\overline{F}$
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	0	1

	F	$\overline{F}$
Minterms (SOP)		
Maxterms (POS)		

One final example

A	B	C	F	$\overline{F}$	m/M
0	0	0	0	1	0
0	0	1	1	0	1
0	1	0	0	1	2
0	1	1	1	0	3
1	0	0	0	1	4
1	0	1	1	0	5
1	1	0	1	0	6
1	1	1	0	1	7

F

$\overline{F}$

Sum of products
(SOP)

$$\Sigma m(1,3,5,6)$$

$$\Sigma m(0,2,4,7)$$

Product of sums
(POS)

$$\Pi M(0,2,4,7)$$

$$\Pi M(1,3,5,6)$$

Converting between canonical forms

	F	$\overline{F}$
Minterms (SOP)	$\sum m(F = 1)$	$\sum m(F = 0)$
Maxterms (POS)	$\prod M(F = 0)$	$\prod M(F = 1)$

DeMorgans: same terms

$$\overline{\sum m(F = 1)} = \prod M(F = 1)$$

Simplification Methods

Simplification Example

- Simplify $F = XY + X\bar{Y} + \bar{X}Y$

X	Y	F	m
0	0	0	$\bar{X}\bar{Y}$
0	1	1	$\bar{X}Y$
1	0	1	$X\bar{Y}$
1	1	1	XY

- $F = X(Y + \bar{Y}) + \bar{X}Y = X + \bar{X}Y$

- Can this be simplified further?

Simplification example cont'd

- $F = X + \bar{X}Y$
- Recall important identity: $X + XY = X$ or (flipped around) $X = X + XY$
 - so $F = X + \bar{X}Y = (X + XY) + \bar{X}Y = X + XY + \bar{X}Y = X + (X + \bar{X})Y = X + Y$
- so most simplified, $F = X + Y$

The point: simplification not
always so easy / obvious
Additional tools are needed!

X	Y	F	m
0	0	0	$\bar{X}\bar{Y}$
0	1	1	$\bar{X}Y$
1	0	1	$X\bar{Y}$
1	1	1	XY

Karnaugh Maps

Karnaugh Maps (K-Maps)

- K-maps are a nice structure to provide simplified sum-of-product (or product-of-sums) solutions
 - Not necessarily most simplified, but it's good enough (for this course)
- We will K-maps to provide simplified SoP form of functions with up to 4 inputs

Karnaugh maps (a.k.a., K-maps)

- All functions can be expressed with a K-map
- There is one square in the map for each minterm in a function's truth table

X	Y	F
0	0	m0
0	1	m1
1	0	m2
1	1	m3

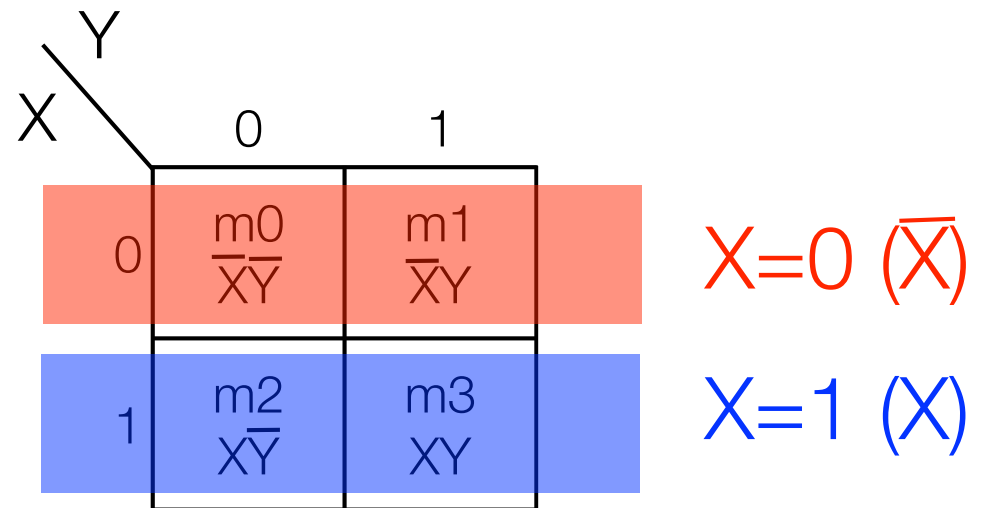
		Y	
		0	1
X	0	m0 $\overline{X}\overline{Y}$	m1 $\overline{X}Y$
	1	m2 $X\overline{Y}$	m3 XY

A K-map is just a 2-dimensional way of representing the function in a truth table

Karnaugh maps

- All functions can be expressed with a map
- There is one square in the map for each minterm in a function's truth table

X	Y	F
0	0	m0
0	1	m1
1	0	m2
1	1	m3

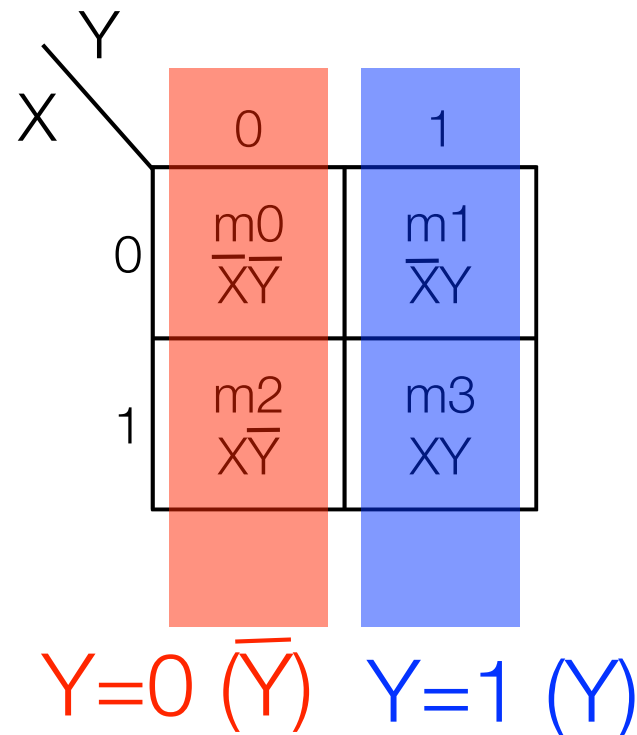


A K-map is just a 2-dimensional way of representing the function in a truth table

Karnaugh maps

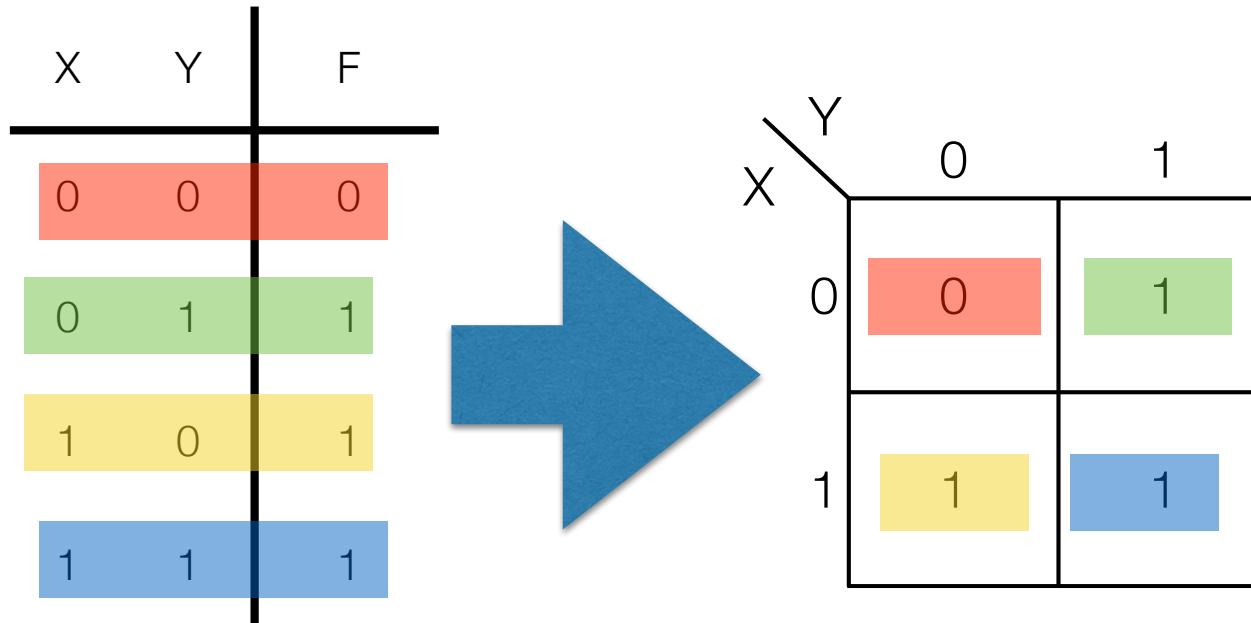
- All functions can be expressed with a map
- There is one square in the map for each minterm in a function's truth table

X	Y	F
0	0	m0
0	1	m1
1	0	m2
1	1	m3



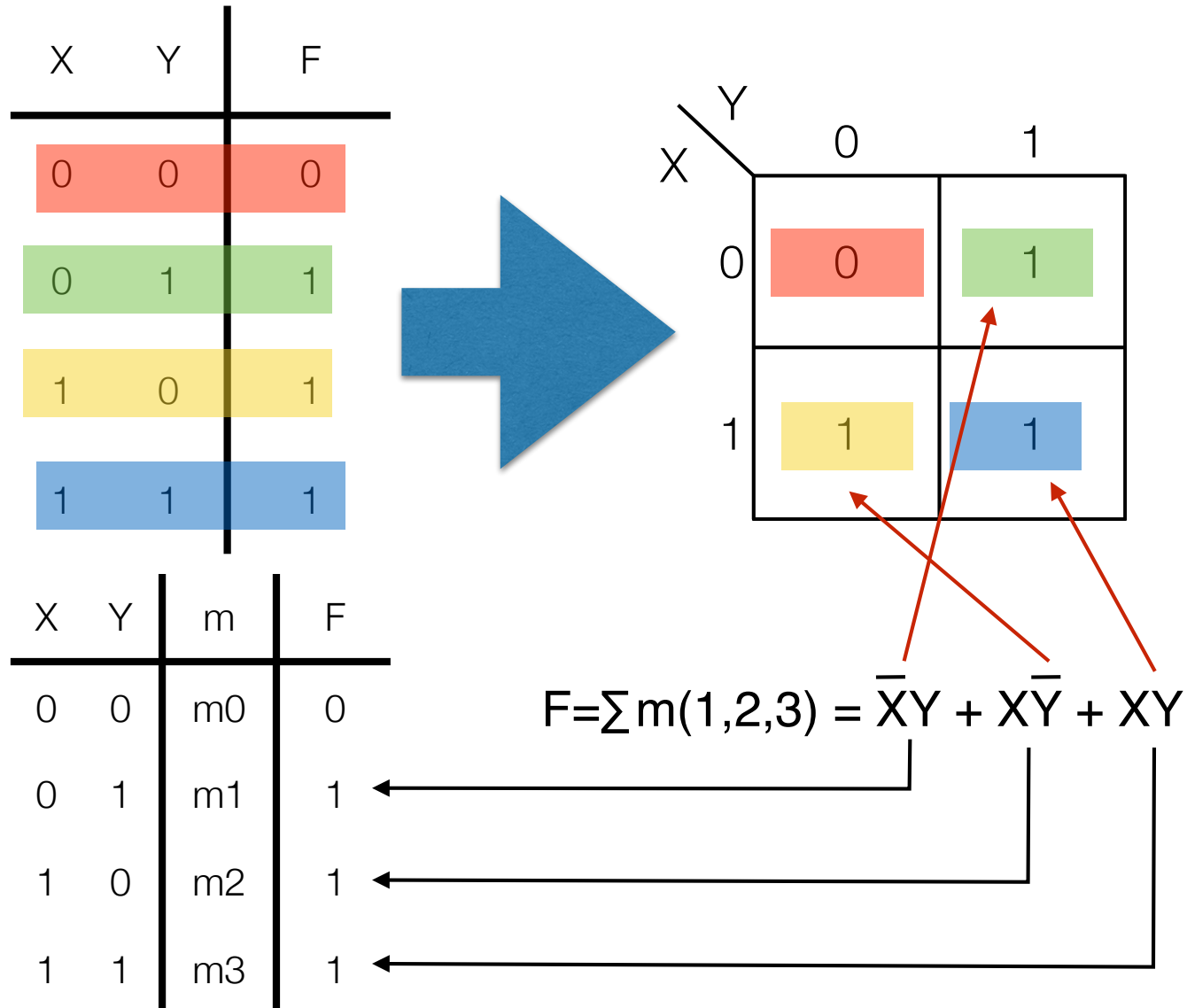
Karnaugh maps express functions

- Fill out table with value of a function



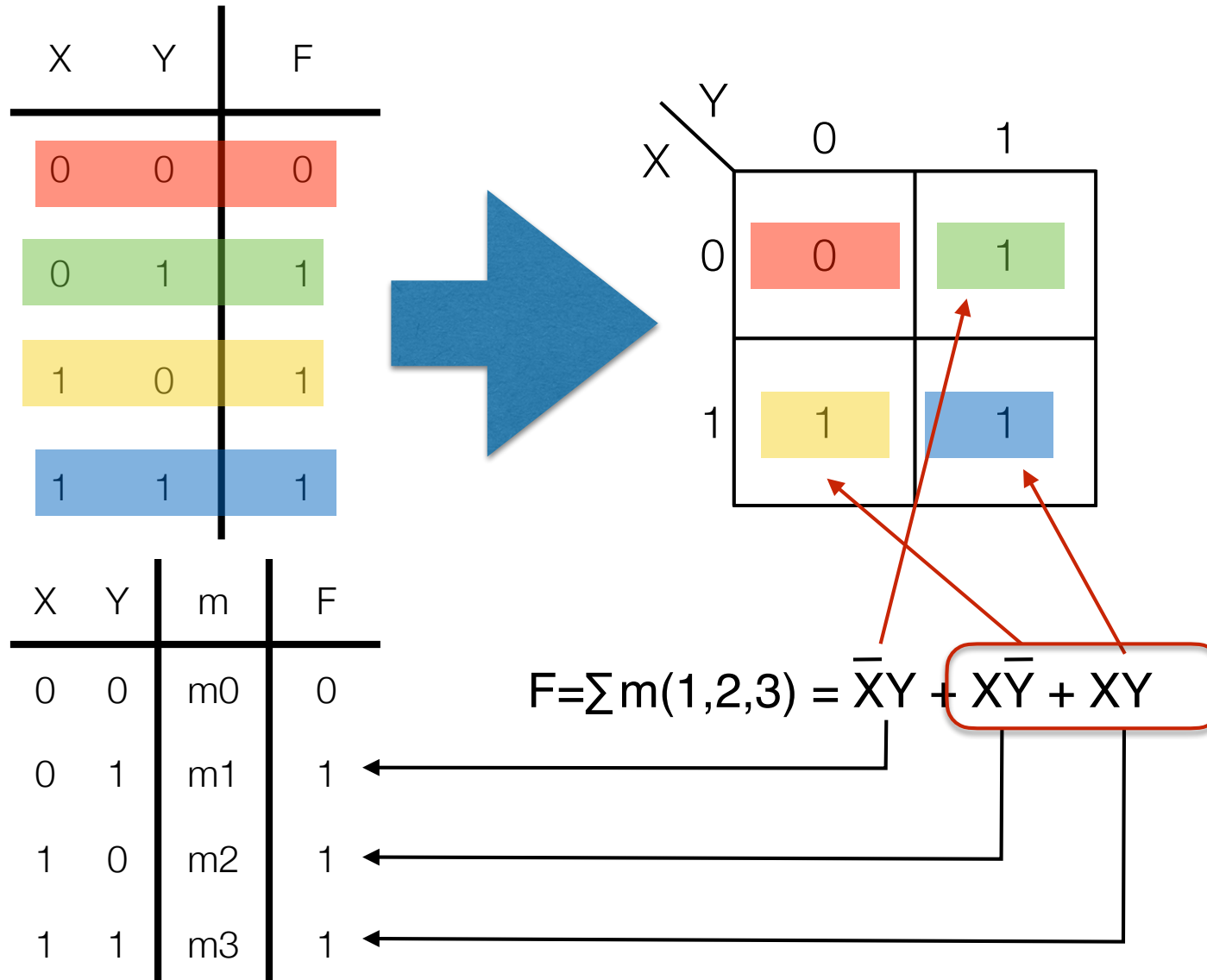
Karnaugh maps express functions

- Fill out table with value of a function



Karnaugh maps express functions

- Fill out table with value of a function

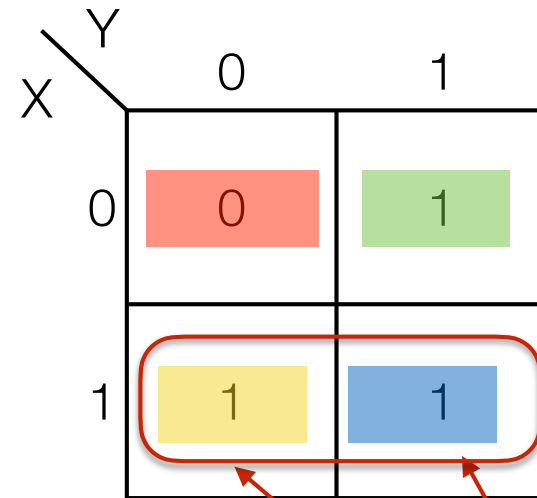


Karnaugh maps express functions

- Fill out table with value of a function

$$X\bar{Y} + XY = X$$

To simplify, when both minterms XY and $X\bar{Y}$ evaluate to 1, they can be “replaced” in the expression by X



X	Y	m	F
0	0	m0	0
0	1	m1	1
1	0	m2	1
1	1	m3	1

$$F = \sum m(1,2,3) = \bar{X}\bar{Y} + X\bar{Y} + XY$$

Simplification using a k-map

- Whenever two squares share an edge and both are 1, those two terms can be combined to form a single term with one less variable

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = \bar{X}\bar{Y} + X\bar{Y} + XY$$

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = X + \bar{X}Y$$

(combined $X\bar{Y} + XY = X$)

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = Y + X\bar{Y}$$

(combined $\bar{X}Y + XY = Y$)

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = X + Y$$

(combined $X\bar{Y} + XY = X$ and $\bar{X}Y + XY = Y$)

Simplification using a k-map

- Whenever two squares share an edge and both are 1, those two terms can be combined to form a single term with one less variable

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = \bar{X}\bar{Y} + X\bar{Y} + XY$$

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = X + \bar{X}Y$$

(combined $X\bar{Y} + XY = X$)

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = Y + X\bar{Y}$$

(combined $\bar{X}Y + XY = Y$)

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = X + Y$$

(combined $X\bar{Y} + XY = X$ and $\bar{X}Y + XY = Y$)

(= $X\bar{Y} + XY + \bar{X}Y + XY$): XY term "repeated" to obtain simplified form

Simplification using a k-map (2)

- “Circle” contiguous squares of 1s (# of squares covered must be a power of 2)
- There is a correspondence between circles on a k-map and product terms in a function expression
- The bigger the circle, the simpler the term
- Add circles (and terms) until all 1s on the k-map are circled

		Y	
		0	1
X	0	0	1
	1	1	1

$$F = X + Y$$

SoP / minterm recap

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z
- e.g., when $W=0,X=1,Y=1,Z=0$, then $F=1$ ($F(0,1,1,0) = 1$)

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z
- e.g., when $W=0,X=1,Y=1,Z=0$, then $F=1$ ($F(0,1,1,0) = 1$)
- When represented as sum of minterms, will include term $\overline{W}XY\overline{Z}$
 - $F = \text{yada yada} + \overline{W}XY\overline{Z}$ will evaluate to 1 when $W=0,X=1,Y=1,Z=0$

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z
- e.g., when $W=0,X=1,Y=1,Z=1$, then $F=0$
- When represented as sum of minterms, F **cannot** include $\overline{W}XYZ$
- because yada yada + $\overline{W}XYZ$ will evaluate to 1 when $W=0,X=1,Y=1,Z=1$

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- As sum of minterms:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$

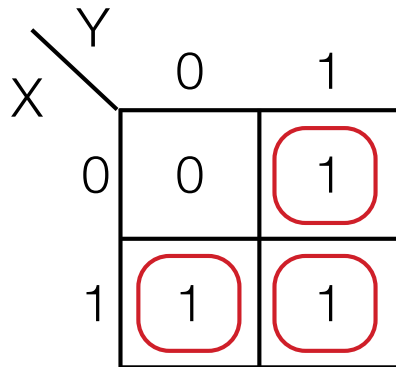
K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

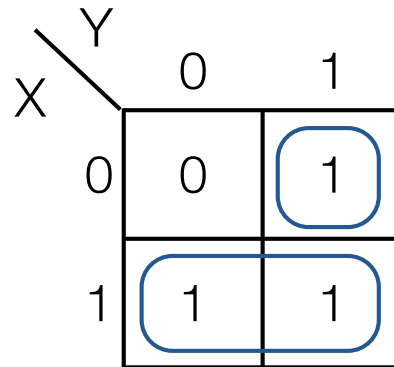
- As sum of minterms:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$
- But maybe we can reduce further by combining minterms to form product terms with fewer variables
 - e.g., $\bar{W}\bar{X}Y\bar{Z} + W\bar{X}Y\bar{Z} = \bar{X}Y\bar{Z}$
 - Can reduce two 4-literal terms with a single 3-literal term!
- Q: How do we find the “best” set of product terms that “cover” all the minterms we need to include to describe our function?
- A: K-map

Recall: Simplification using a 2-variable k-map

- Whenever two squares share an edge and both are 1, those two terms can be combined to form a single term with one less variable

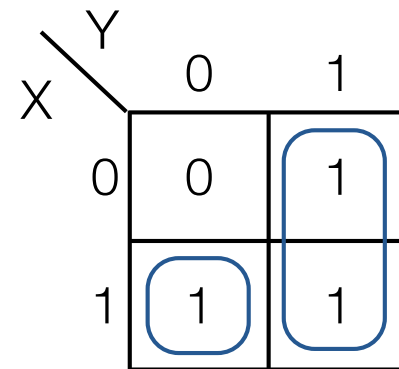


$$F = \bar{X}Y + X\bar{Y} + XY$$



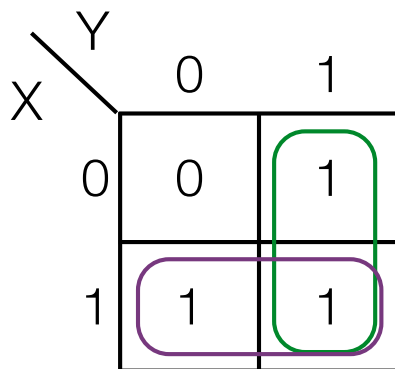
$$F = X + \bar{X}Y$$

(combined $X\bar{Y} + XY = X$)



$$F = Y + X\bar{Y}$$

(combined $\bar{X}Y + XY = Y$)



$$F = X + Y$$

(combined $X\bar{Y} + XY = X$ and $\bar{X}Y + XY = Y$)

Best Option!

(= $X\bar{Y} + XY + \bar{X}Y + XY$): XY term “repeated” to obtain simplified form

3-variable Karnaugh maps

- Use gray ordering on edges with multiple variables
- Gray encoding: order of values such that only one bit changes at a time
- Two minterms are considered adjacent if they differ in only one variable (this means maps “wrap”)

		Y Z			
		0 0	0 1	1 1	1 0
X	0	m0 $\bar{X}\bar{Y}\bar{Z}$	m1 $\bar{X}\bar{Y}Z$	m3 $\bar{X}YZ$	m2 $\bar{X}Y\bar{Z}$
	1	m4 $X\bar{Y}\bar{Z}$	m5 $X\bar{Y}Z$	m7 XYZ	m6 $XY\bar{Z}$

Or just

X

3-variable Karnaugh maps

- Use gray ordering on edges with multiple variables
- Gray encoding: order of values such that only one bit changes at a time
- Two minterms are considered adjacent if they differ in only one variable (this means maps “wrap”)

		Y Z			
X		0 0	0 1	1 1	1 0
	0	m0 $\bar{X}\bar{Y}\bar{Z}$	m1 $\bar{X}\bar{Y}Z$	m3 $\bar{X}YZ$	m2 $\bar{X}Y\bar{Z}$
	1	m4 $X\bar{Y}\bar{Z}$	m5 $X\bar{Y}Z$	m7 XYZ	m6 $XY\bar{Z}$

Or just

$\bar{X}$

3-variable Karnaugh maps

- Use gray ordering on edges with multiple variables
- Gray encoding: order of values such that only one bit changes at a time
- Two minterms are considered adjacent if they differ in only one variable (this means maps “wrap”)

Y Z

X

Y=1

	0 0	0 1	1 1	1 0
0	m0 $\bar{X}\bar{Y}\bar{Z}$	m1 $\bar{X}\bar{Y}Z$	m3 $\bar{X}YZ$	m2 $\bar{X}Y\bar{Z}$
1	m4 $X\bar{Y}\bar{Z}$	m5 $X\bar{Y}Z$	m7 XYZ	m6 $XY\bar{Z}$

Or just

Y

3-variable Karnaugh maps

- Use gray ordering on edges with multiple variables
- Gray encoding: order of values such that only one bit changes at a time
- Two minterms are considered adjacent if they differ in only one variable (this means maps “wrap”)

Y Z

X

Y=0

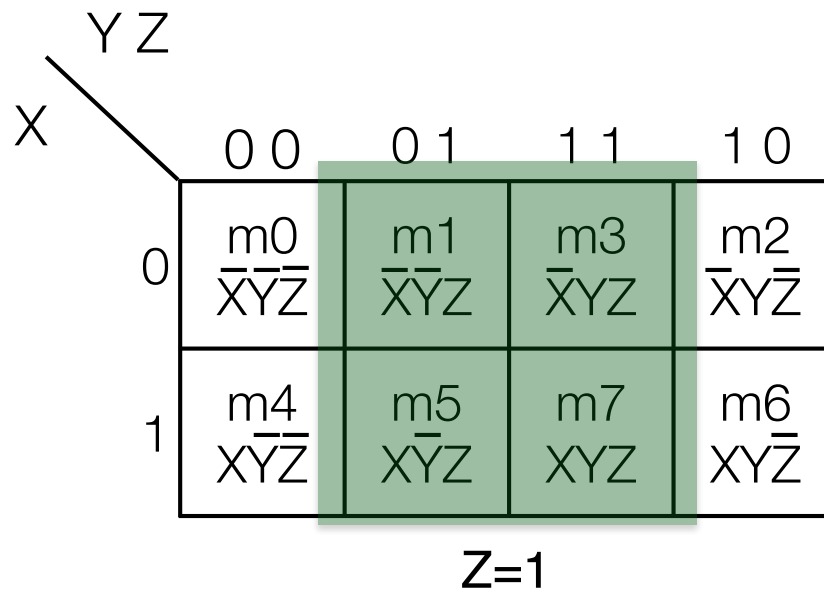
	0 0	0 1	1 1	1 0
0	m0 $\bar{X}\bar{Y}\bar{Z}$	m1 $\bar{X}\bar{Y}Z$	m3 $\bar{X}YZ$	m2 $\bar{X}Y\bar{Z}$
1	m4 $X\bar{Y}\bar{Z}$	m5 $X\bar{Y}Z$	m7 XYZ	m6 $XY\bar{Z}$

Or just

$\bar{Y}$

3-variable Karnaugh maps

- Use gray ordering on edges with multiple variables
- Gray encoding: order of values such that only one bit changes at a time
- Two minterms are considered adjacent if they differ in only one variable (this means maps “wrap”)

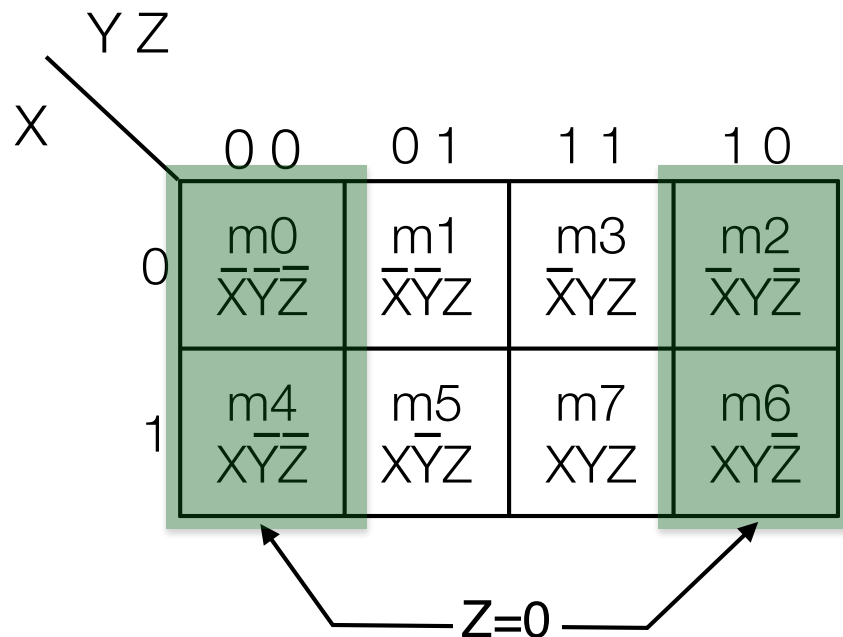


Or just

Z

3-variable Karnaugh maps

- Use gray ordering on edges with multiple variables
- Gray encoding: order of values such that only one bit changes at a time
- Two minterms are considered adjacent if they differ in only one variable (this means maps “wrap”)

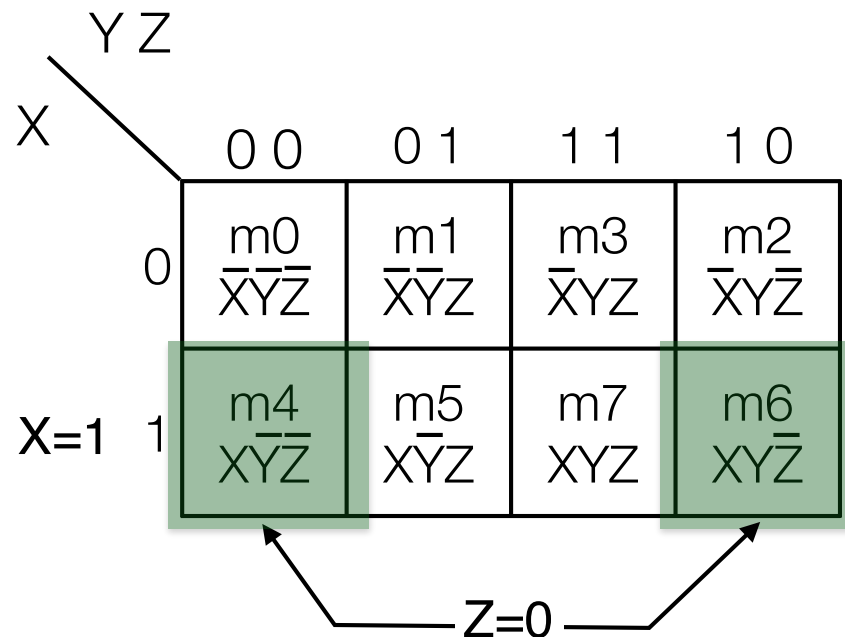


Or just

$\bar{Z}$

3-variable Karnaugh maps

- Use gray ordering on edges with multiple variables
- Gray encoding: order of values such that only one bit changes at a time
- Two minterms are considered adjacent if they differ in only one variable (this means maps “wrap”)



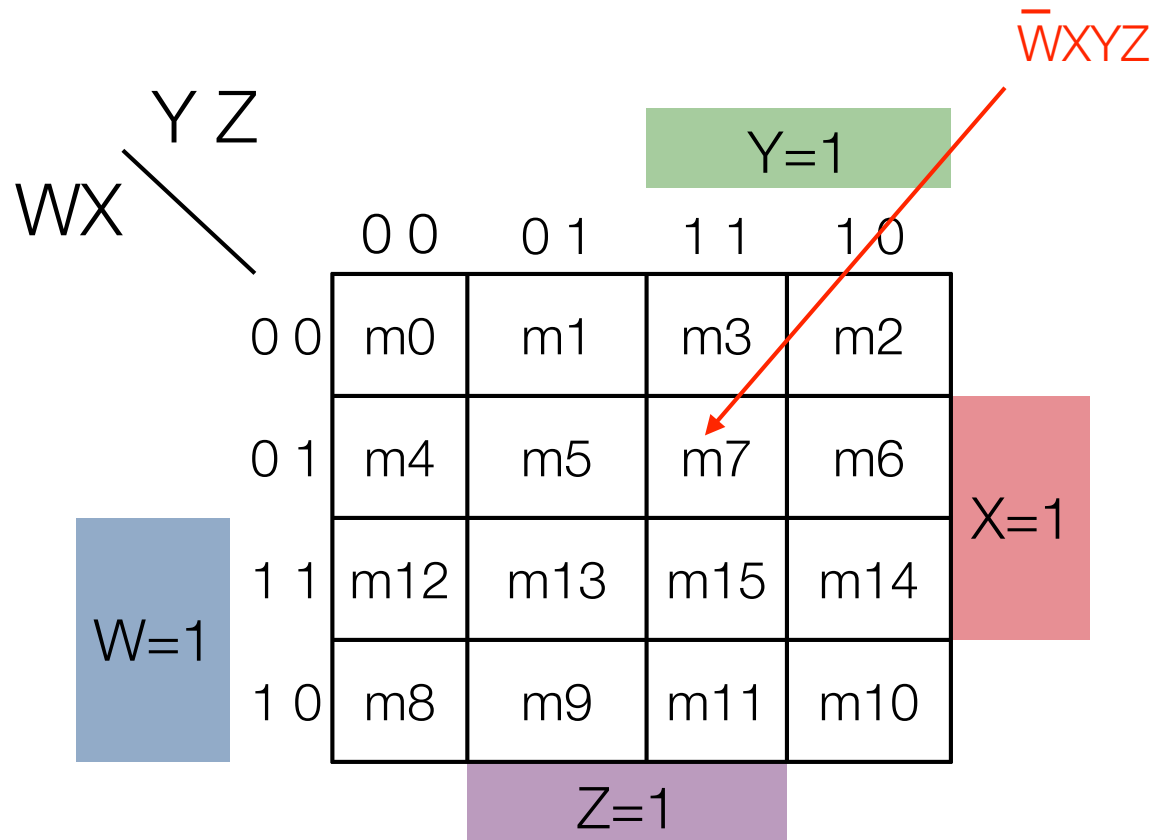
$X=1$ and $Z=0$

Or just

$X\bar{Z}$

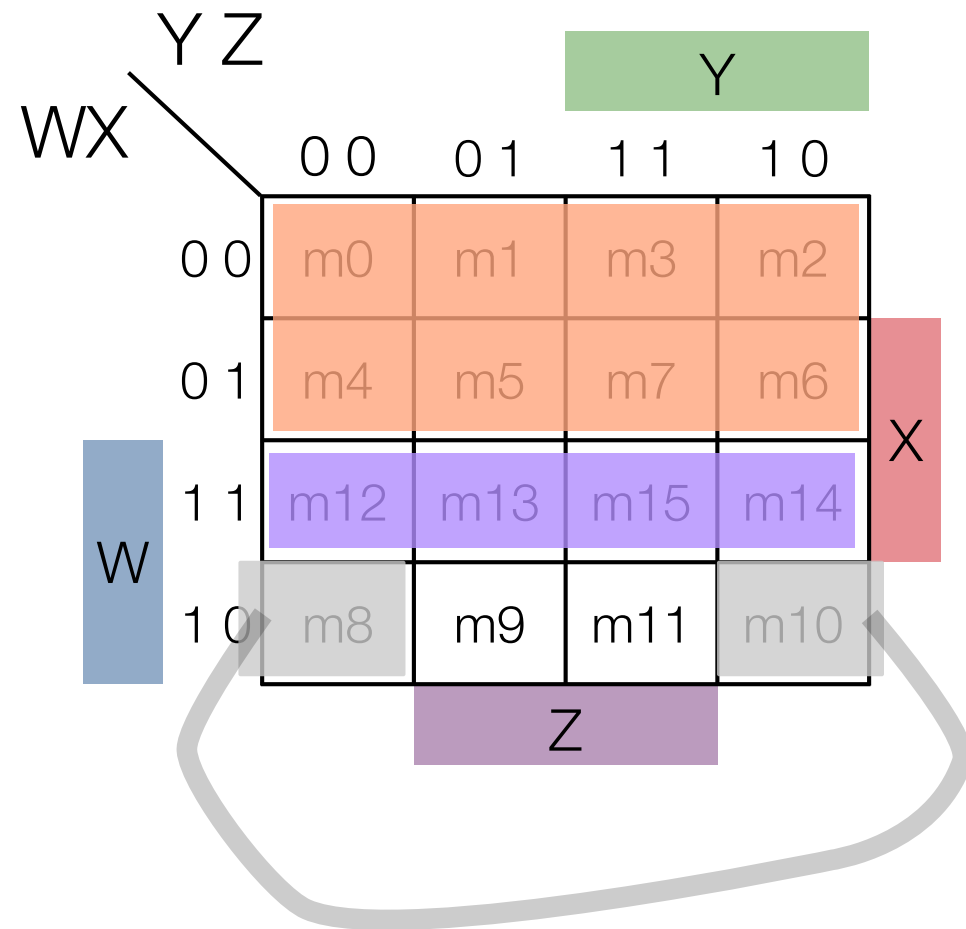
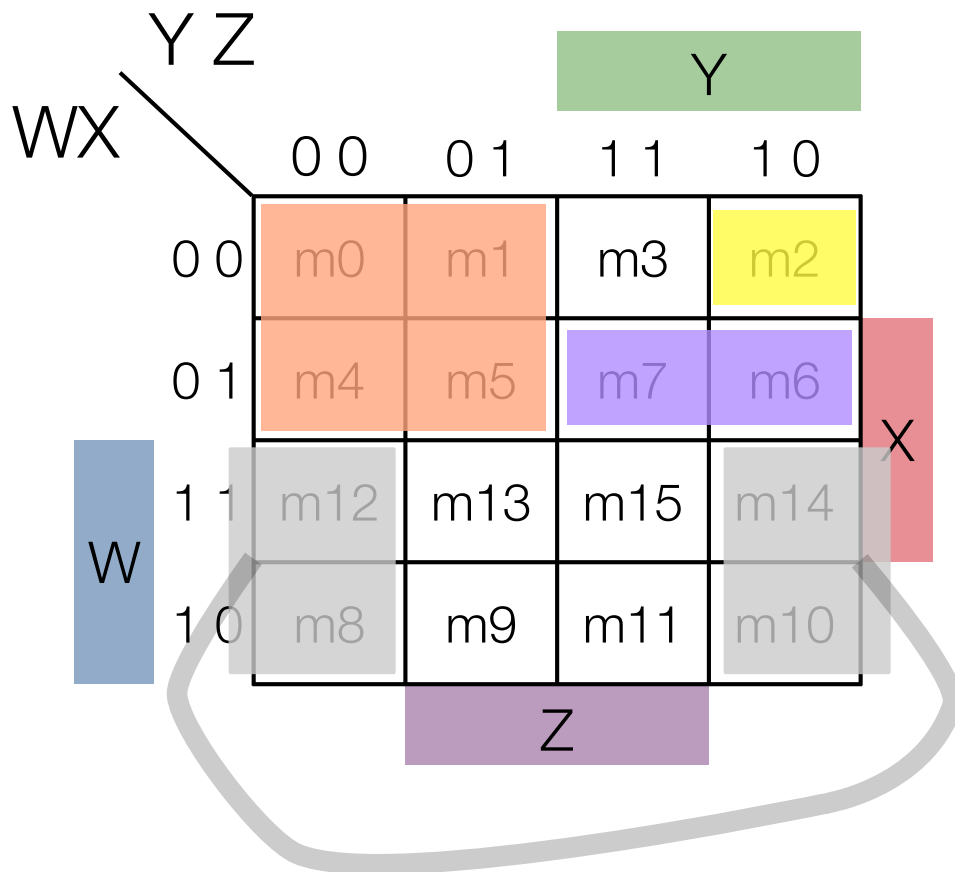
4-variable Karnaugh maps

- Extension of 3-variable maps



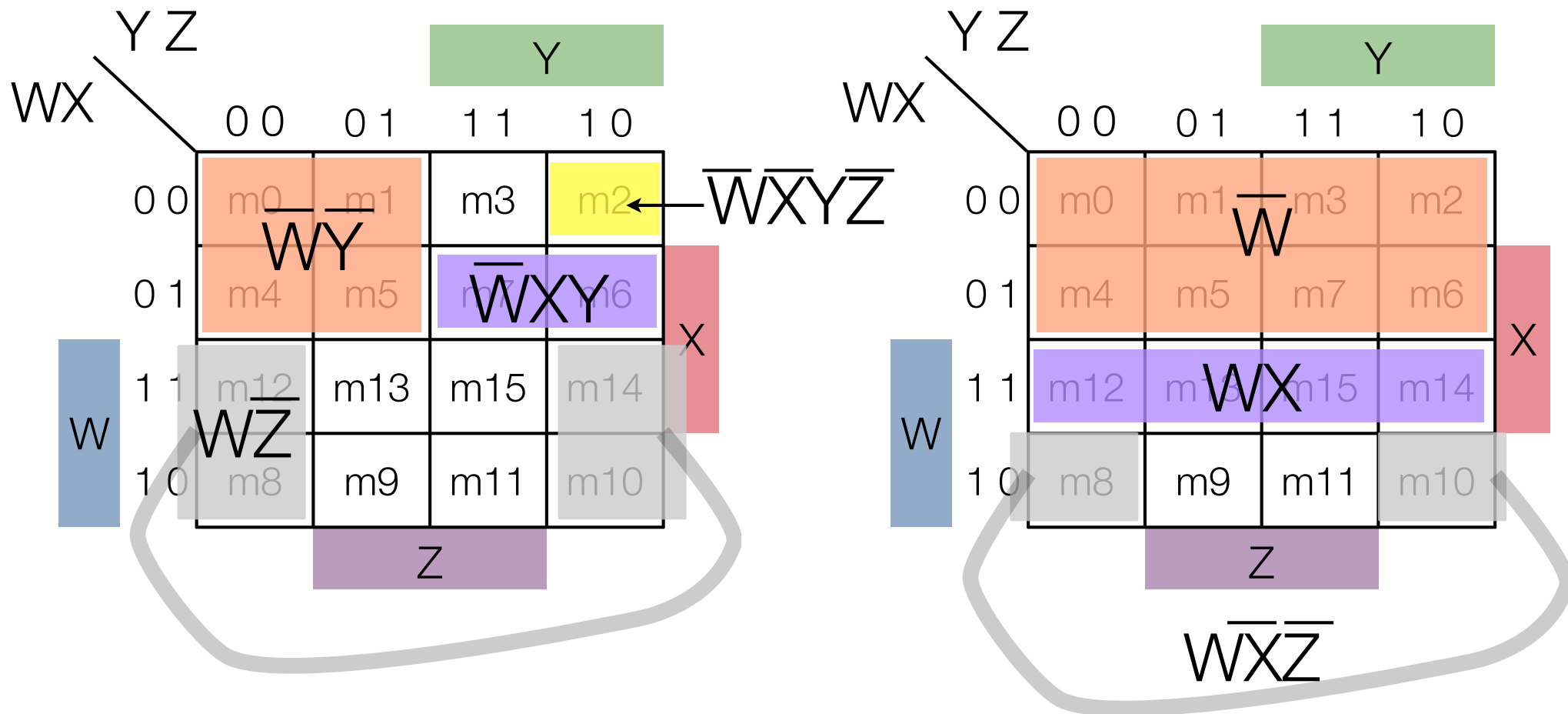
Product terms are just $2^i \times 2^j$ boxes (that might wrap around)

- A **product term**, which, viewed in a K-Map is a $2^i \times 2^j$ size “rectangle” (possibly wrapping around) where $i=0,1,2$, $j=0,1,2$



Product terms are just $2^i \times 2^j$ boxes (that might wrap around)

- A **product term**, which, viewed in a K-Map is a $2^i \times 2^j$ size “rectangle” (possibly wrapping around) where $i=0,1,2$, $j=0,1,2$



Note: bigger rectangles = fewer literals

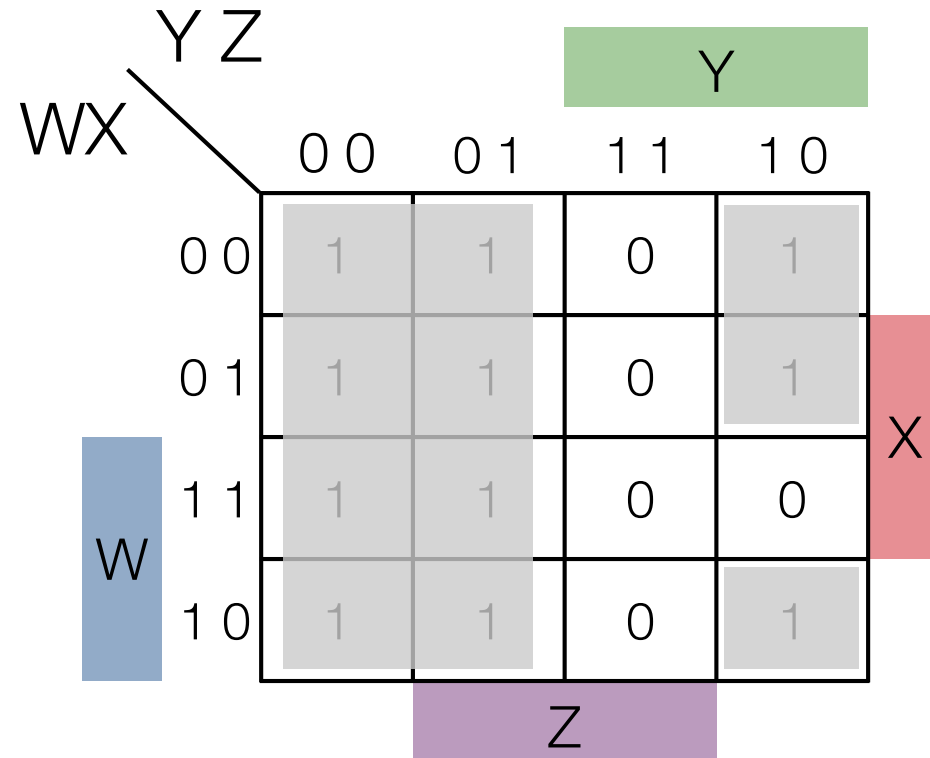
4-variable Karnaugh maps example

W	X	Y	Z	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

		YZ			
		Y 00 01 11 10			
WX	00	1	1	0	1
	01	1	1	0	1
	11	1	1	0	0
	10	1	1	0	1
W		Z			
		X			

4-variable Karnaugh maps example

W	X	Y	Z	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0



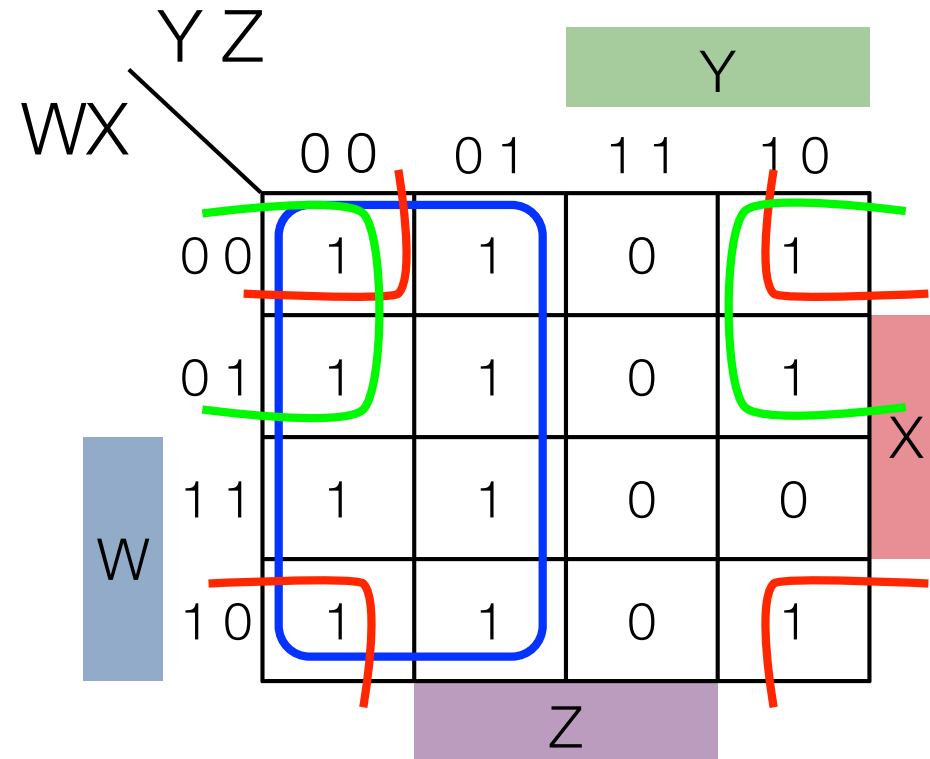
K-maps make F expressed as SoP easy to see, e.g.,

$$\bar{Y} + \bar{W}Y\bar{Z} + W\bar{X}Y\bar{Z}$$

Can the expression for F be simplified further?

4-variable Karnaugh maps example

W	X	Y	Z	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0



$$\bar{Y} + \bar{W}\bar{Z} + \bar{X}\bar{Z}$$

Rule when picking product terms: Must cover only 1's, but OK to overlap. Bigger rectangles are better (fewer literals)

K-Map Recap

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z
- e.g., when $W=0,X=1,Y=1,Z=0$, then $F=1$ ($F(0,1,1,0) = 1$)

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z
- e.g., when $W=0,X=1,Y=1,Z=0$, then $F=1$ ($F(0,1,1,0) = 1$)
- When represented as sum of minterms, will include term $\overline{W}XY\overline{Z}$
 - $F = \text{yada yada} + \overline{W}XY\overline{Z}$ will evaluate to 1 when $W=0,X=1,Y=1,Z=0$

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Start with some arbitrary function (in this case, of 4 variables, $F(W,X,Y,Z)$)
- Function is defined for all possible assignments of $\{0,1\}$ to each of W,X,Y,Z
- e.g., when $W=0,X=1,Y=1,Z=1$, then $F=0$
- When represented as sum of minterms, F **cannot** include $\overline{W}XYZ$
- because yada yada + $\overline{W}XYZ$ will evaluate to 1 when $W=0,X=1,Y=1,Z=1$

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- As sum of minterms:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- As sum of minterms:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$
- On a K-map:

		YZ			
		00	01	11	10
WX	00	0	0	0	1
	01	0	0	0	1
	11	1	1	1	1
	10	1	0	0	1

Y
X
W
Z

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- As sum of minterms:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$
- On a K-map:

		Y Z			
		0 0	0 1	1 1	1 0
W X	0 0	0	0	0	1
	0 1	0	0	0	1
	1 1	1	1	1	1
	1 0	1	0	0	1

Y
X
W
Z

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- As sum of minterms:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$
- Find simplified SoP form without K-map: start trying to “combine” terms
 - e.g., $WXY\bar{Z} + WXYZ = WXY$
 - and $WX\bar{Y}\bar{Z} + WX\bar{Y}Z = WX\bar{Y}$
 - and $WXY + WX\bar{Y} = WX$
- So partially simplifies to:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX$
- keep going... (i.e., top part simplifies)

K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- As sum of minterms:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$
- Partially simplified as:
- $F = \bar{W}\bar{X}Y\bar{Z} + \bar{W}XY\bar{Z} + W\bar{X}\bar{Y}\bar{Z} + W\bar{X}Y\bar{Z} + WX$
 - WX is a product term but not a minterm
 - It “replaced” the whole bottom row
 - WX “covers” 4 minterms
 - Easy to spot in the truth table... it’s the last 4 rows

K-map Recap

- Let's simplify using a K-map:
- Recall: each possible product term is a $2^i \times 2^j$ “rectangle” (possibly wrapping around)
- 3rd K-map row means solution must include minterms:
 - $WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$

		Y Z			
		0 0	0 1	1 1	1 0
W X	0 0	0	0	0	1
	0 1	0	0	0	1
	1 1	1	1	1	1
	1 0	1	0	0	1

K-map Recap

- Let's simplify using a K-map:
- Recall: each possible product term is a $2^i \times 2^j$ “rectangle” (possibly wrapping around)
- 3rd K-map row means solution must include minterms:
 - $WX\bar{Y}\bar{Z} + WX\bar{Y}Z + WXY\bar{Z} + WXYZ$
- Better: Could replace with:
 - $WX\bar{Y} + WXY$
- Even better: just WX
- WX is a prime implicant, can't replace by W or X (there are 0's)

		Y Z			
		0 0	0 1	1 1	1 0
W X	0 0	0	0	0	1
	0 1	0	0	0	1
	1 1	1	1	1	1
	1 0	1	0	0	1

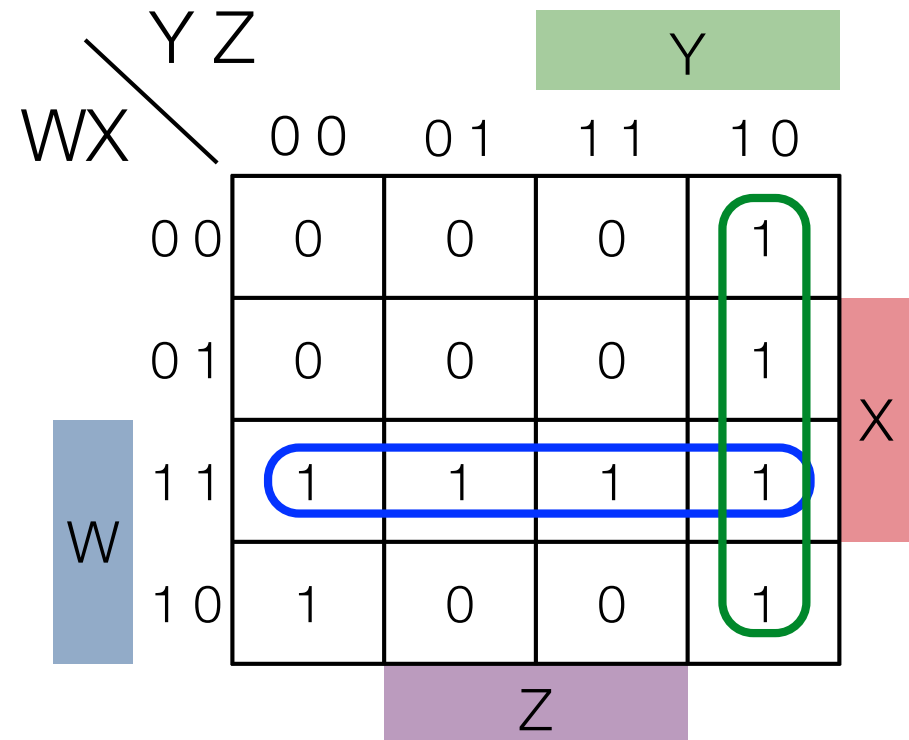
K-map Recap

W	X	Y	Z	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- $Y\bar{Z}$ is another product term we could use...
- Harder to spot on the Truth Table: rows are “spread out”

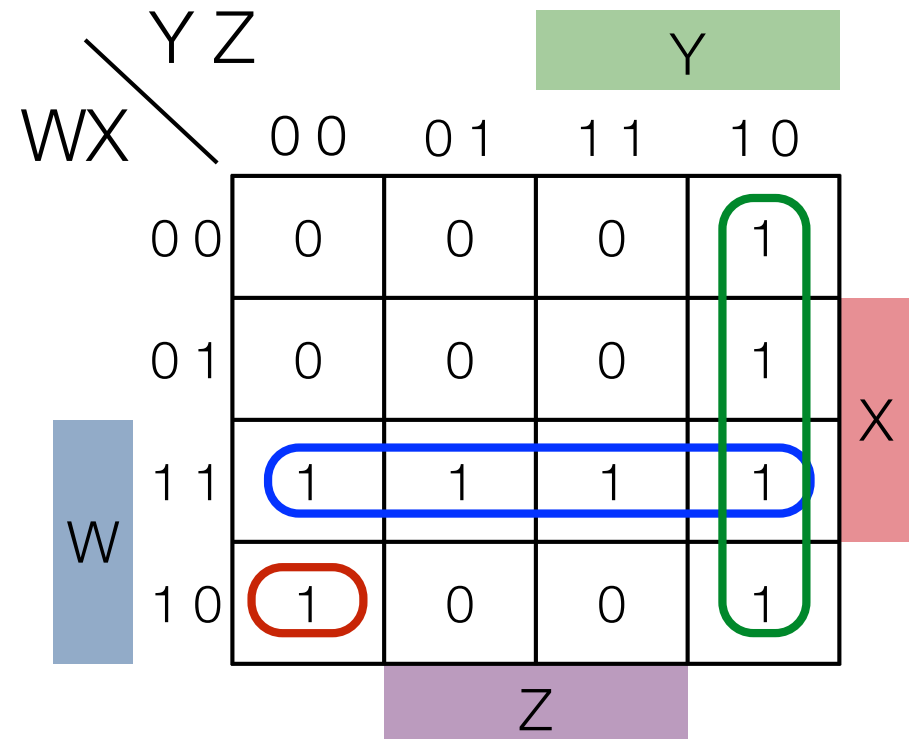
K-map Recap

- $WX + Y\bar{Z}$: overlap is good! (double-counting minterms OK)
- Still have one more 1 to cover (bottom left):
 - right now, $F = WX + Y\bar{Z}$ evaluates correctly except for $W=1, X=0, Y=0, Z=0$



K-map Recap

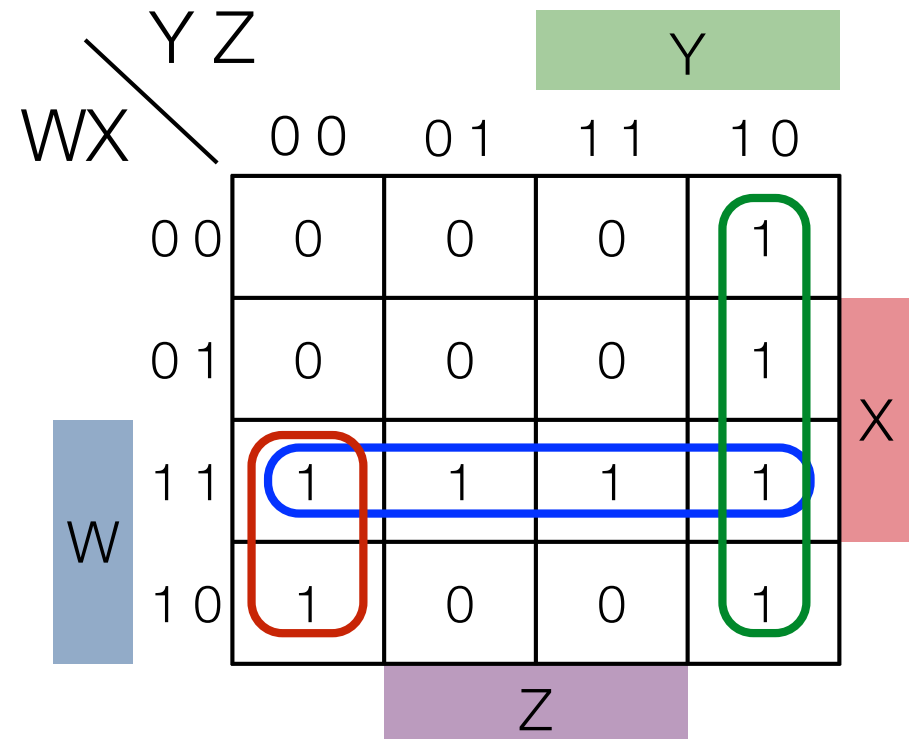
- $WX + Y\bar{Z}$: overlap is good! (double-counting minterms OK)
- Still have one more 1 to cover (bottom left):
 - right now, $F = WX + Y\bar{Z}$ evaluates correctly except for $W=1, X=0, Y=0, Z=0$
 - Could just add minterm so $F = WX + Y\bar{Z} + W\bar{X}\bar{Y}\bar{Z}$



K-map Recap

- $WX + Y\bar{Z}$: overlap is good! (double-counting minterms OK)
- Still have one more 1 to cover (bottom left):
 - right now, $F = WX + Y\bar{Z}$ evaluates correctly except for $W=1, X=0, Y=0, Z=0$
 - Could just add minterm so $F = WX + Y\bar{Z} + W\bar{X}\bar{Y}\bar{Z}$

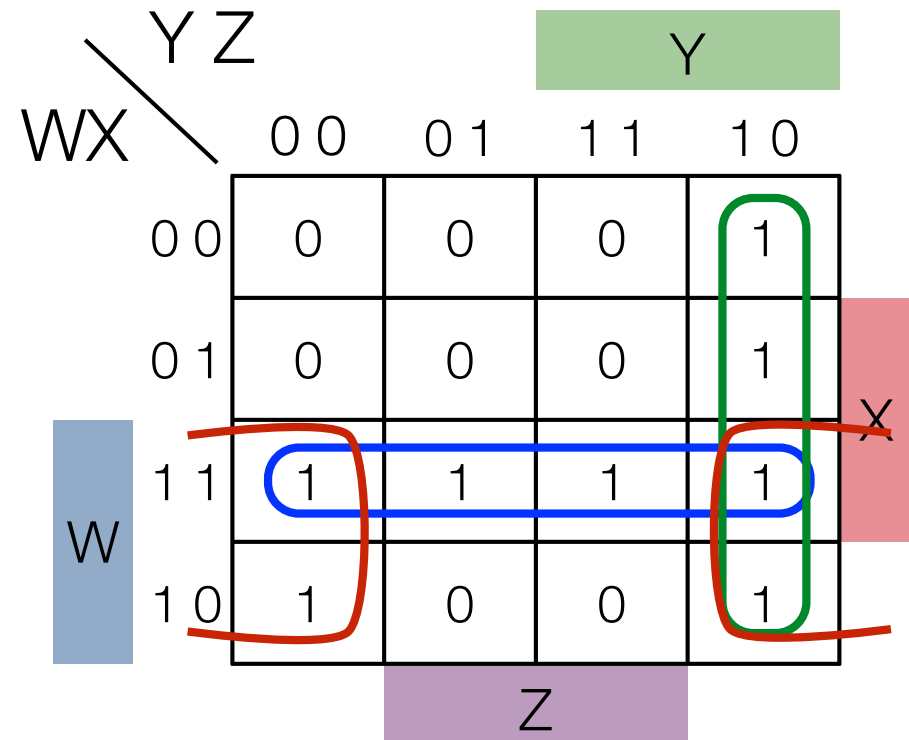
- Better: Use $W\bar{Y}\bar{Z}$
- so $F = WX + Y\bar{Z} + W\bar{Y}\bar{Z}$



K-map Recap

- $WX + Y\bar{Z}$: overlap is good! (double-counting minterms OK)
- Still have one more 1 to cover (bottom left):
 - right now, $F = WX + Y\bar{Z}$ evaluates correctly except for $W=1, X=0, Y=0, Z=0$
 - Could just add minterm so $F = WX + Y\bar{Z} + W\bar{X}\bar{Y}\bar{Z}$

- Better: Use $W\bar{Y}\bar{Z}$
- so $F = WX + Y\bar{Z} + W\bar{Y}\bar{Z}$
- Best - use $W\bar{Z}$
- so $F = WX + Y\bar{Z} + W\bar{Z}$
- K-map shows us this is best (for SoP)



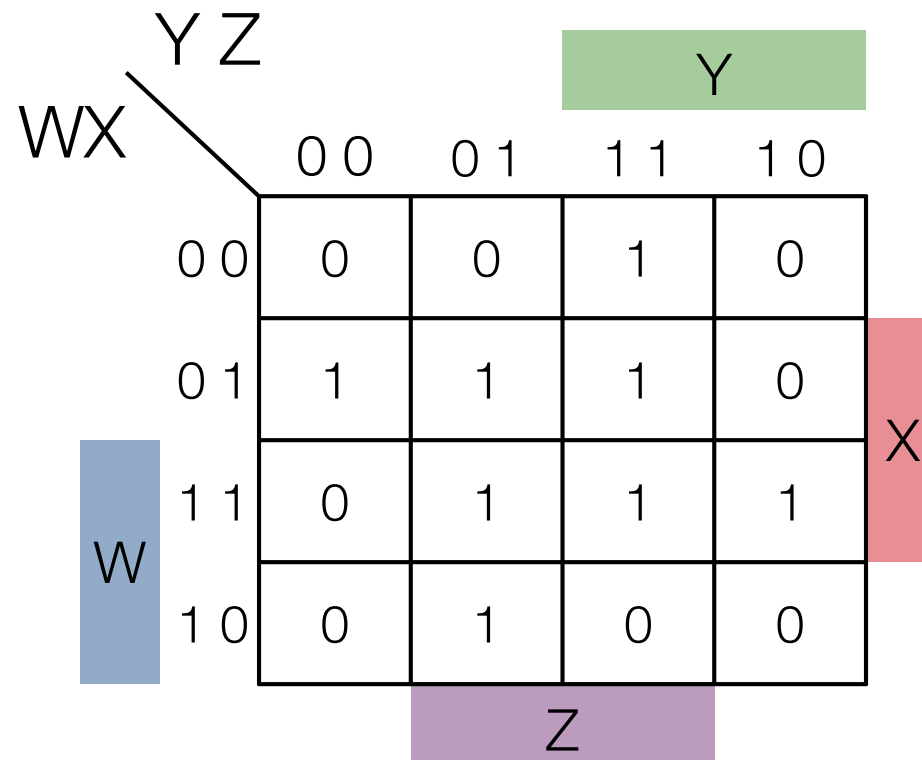
Back to our regularly
scheduled
programming...

Implicant terminology for a function F

- Given a function F :
 - An **implicant** is a product term (i.e., a “box” whose dimensions are powers of 2) comprised of minterms for which the function F evaluates to 1
 - i.e., the “box” must only cover squares that are 1
- **prime implicant**: An implicant not contained within another implicant (remember that implicant dimensions must be powers of 2!)
- **essential prime implicant**: a **prime implicant** that is the **only prime implicant** to cover some minterm.

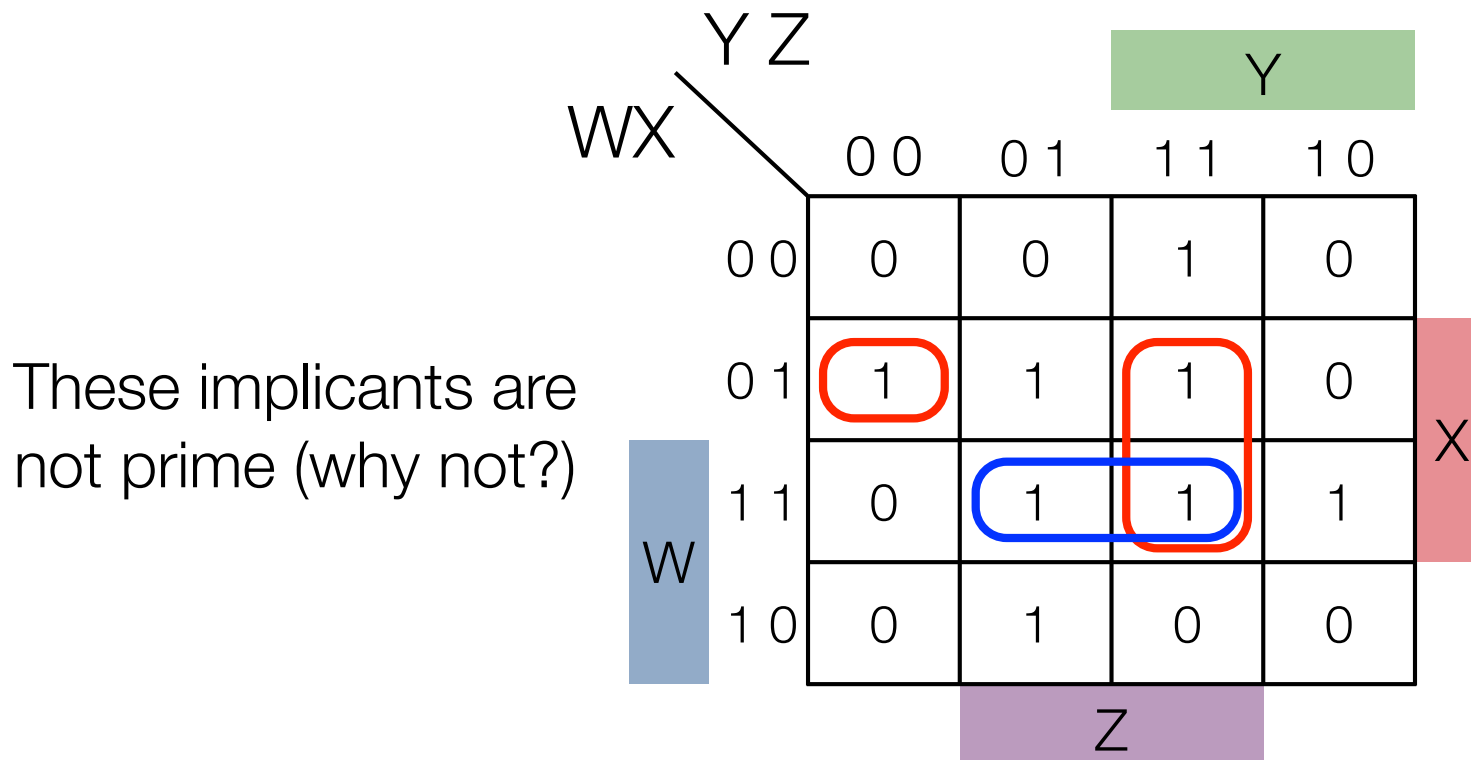
4-variable Karnaugh maps (3)

- List all of the prime implicants for this function
- Is any of them an essential prime implicant?
- What is a simplified expression for this function?



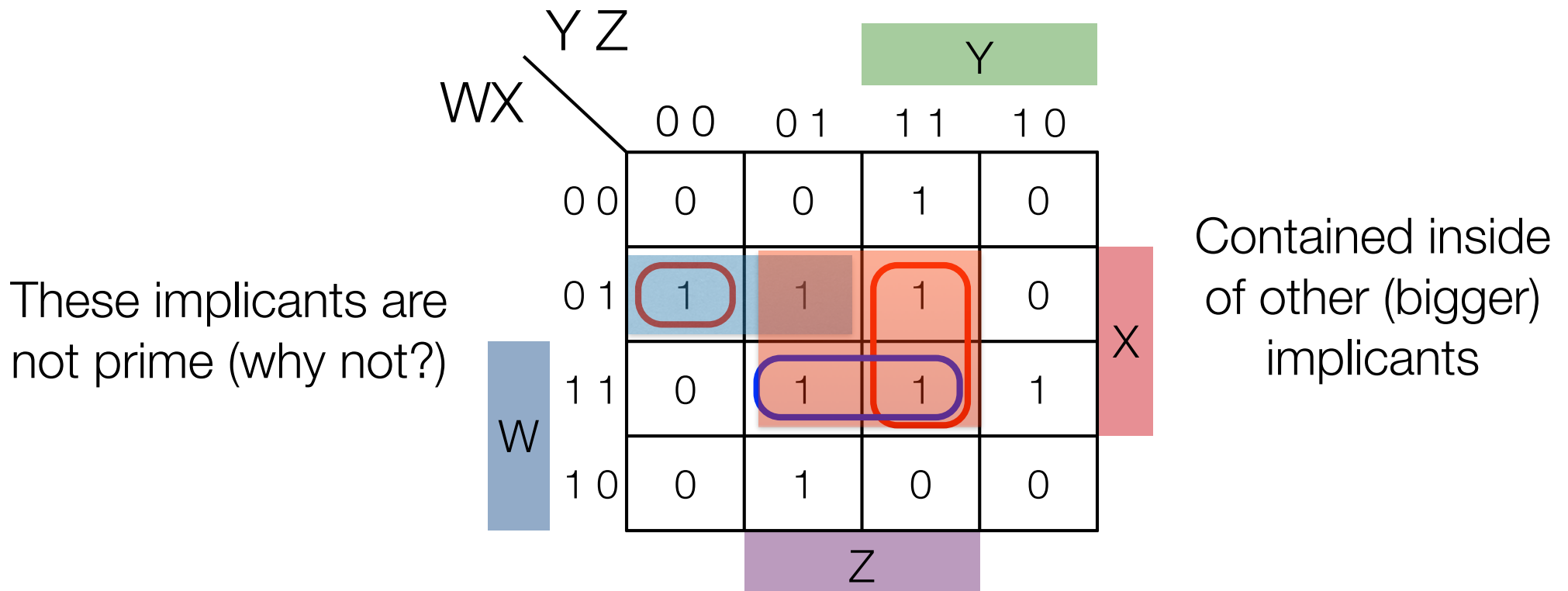
4-variable Karnaugh maps (3)

- List all of the prime implicants for this function
- Is any of them an essential prime implicant?
- What is a simplified expression for this function?



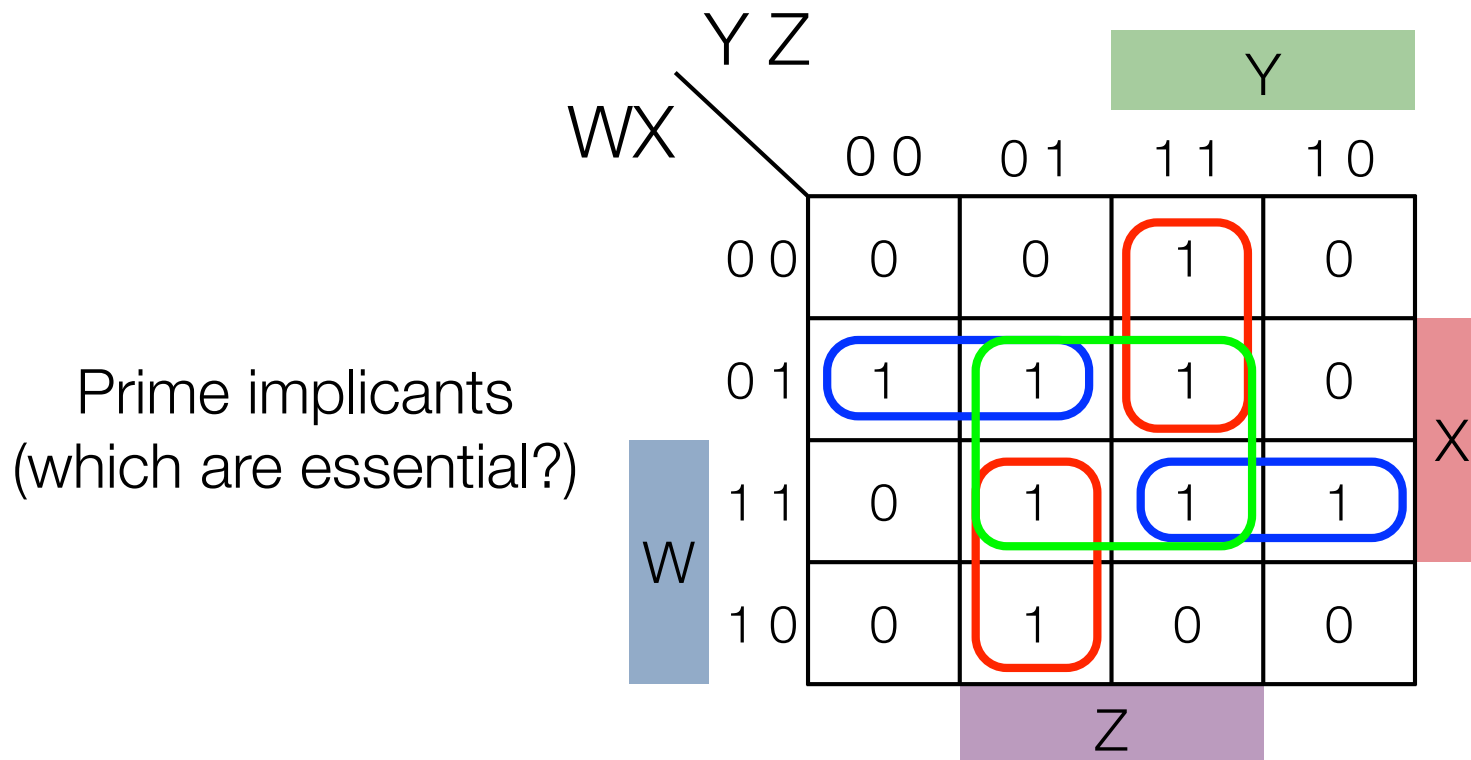
4-variable Karnaugh maps (3)

- List all of the prime implicants for this function
- Is any of them an essential prime implicant?
- What is a simplified expression for this function?



4-variable Karnaugh maps (3)

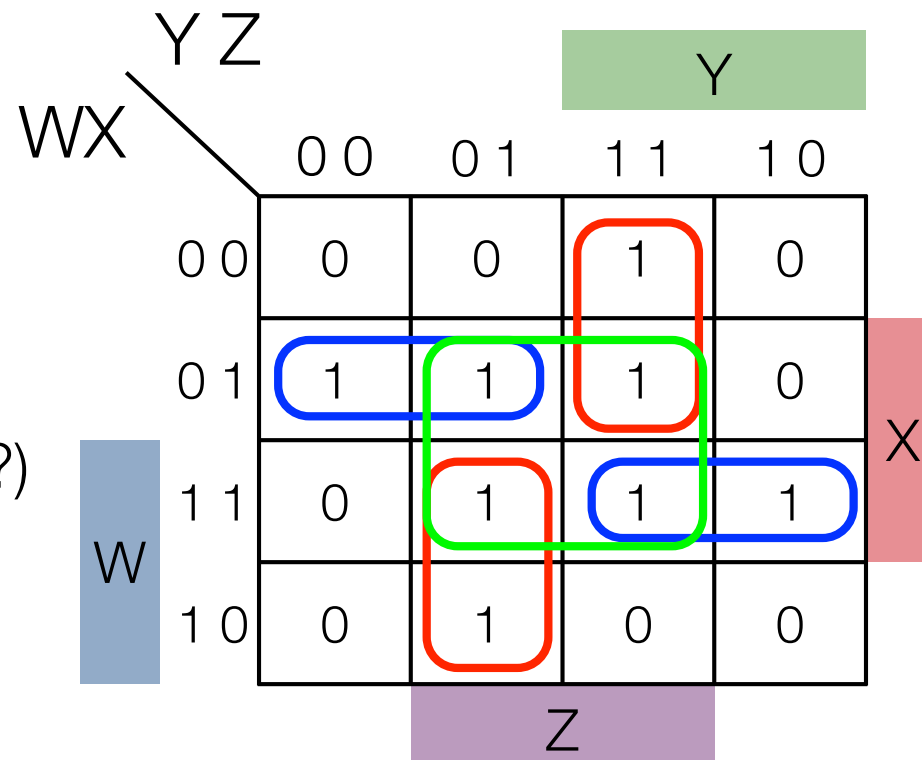
- List all of the prime implicants for this function
- Is any of them an essential prime implicant?
- What is a simplified expression for this function?



4-variable Karnaugh maps (3)

- List all of the prime implicants for this function
- Is any of them an essential prime implicant?
- What is a simplified expression for this function?

Prime implicants
(which are essential?)



The blue and red are essential: each has 1 minterm not covered by any other prime implicant

Green not essential: every covered minterm also covered by blue or red prime implicant

Using K-maps to build simplified circuits

- Step 1: Identify all PIs and essential PIs
- Step 2: Include all Essential PIs in the circuit (Why?)
- Step 3: If any 1-valued minterms are uncovered by EPIs, choose PIs that are “big” and do a good job covering
 - **Selection Rule:** a heuristic for usually choosing “good” PIs: choose the PIs that minimize overlap with one another and with EPIs

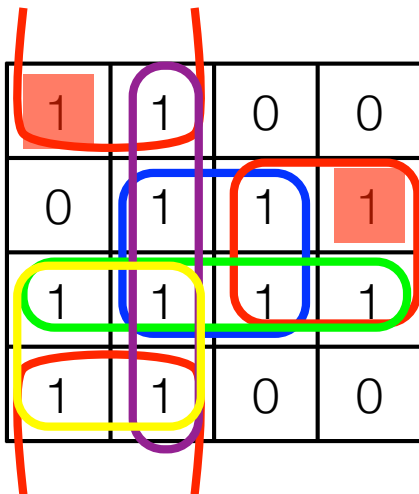
1	1	1	0
0	1	1	0
1	1	1	1
1	1	0	1

1	1	0	0
0	1	1	0
0	0	1	1
1	0	0	1

Using K-maps to build simplified circuits

- Step 1: Identify all PIs and essential PIs
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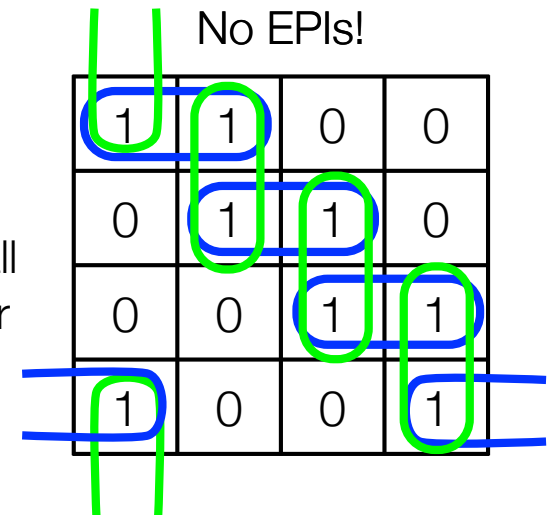
Red bounds are EPIs (solo-covered minterm shown in red)



Also need
(purple or blue) and
(yellow or green)

No EPIs!

All blue PIs or all
green PIs cover



Useful(?) function example

Design example : 2-bit multiplier

a ₁	a ₀	b ₁	b ₀	z ₃	z ₂	z ₁	z ₀
0	0	0	0				
0	0	0	1				
0	0	1	0				
0	0	1	1				
0	1	0	0				
0	1	0	1				
0	1	1	0				
0	1	1	1				
1	0	0	0				
1	0	0	1				
1	0	1	0				
1	0	1	1				
1	1	0	0				
1	1	0	1				
1	1	1	0				
1	1	1	1				

two 2-bit #'s multiplied together to give a 4-bit solution

e.g., $a_1a_0 = 10$, $b_1b_0 = 11$, $z_3z_2z_1z_0 = 0110$
 $2 \quad * \quad 3 \quad = \quad 6$

Design example : 2-bit multiplier (SOLUTION)

a1	a0	b1	b0	z3	z2	z1	z0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

Design example : 2-bit multiplier (SOLUTION)

a1	a0	b1	b0	z3	z2	z1	z0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

			b ₀	
	0	0	0	0
	0	0	0	0
a ₁	0	0	1	0
	0	0	0	0
			b ₁	a ₀

$$z_3 = a_1 a_0 b_1 b_0$$

$$z_1 = (\text{exercise})$$

			b ₀	
	0	0	0	0
	0	0	1	1
a ₁	0	1	0	1
	0	1	1	0
			b ₁	a ₀

			b ₀	
	0	0	0	0
	0	0	0	0
a ₁	0	0	0	1
	0	0	1	1
			b ₁	a ₀

$$z_2 = a_1 \bar{a}_0 b_1 + a_1 b_1 \bar{b}_0$$

$$z_0 = a_0 b_0$$

			b ₀	
	0	0	0	0
	0	1	1	0
a ₁	0	1	1	0
	0	0	0	0
			b ₁	a ₀

Design example : 2-bit multiplier (SOLUTION)

a1	a0	b1	b0	z3	z2	z1	z0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

			b ₀	
	0	0	0	0
	0	0	0	0
	0	0	1	0
a ₁	0	0	0	0
			b ₁	

$$z_3 = a_1 a_0 b_1 b_0$$

$z_1 = \text{(exercise)}$

			b ₀	
	0	0	0	0
	0	0	1	1
	0	1	0	1
a ₁	0	1	1	0
			b ₁	

			b ₀	
	0	0	0	0
	0	0	0	0
	0	0	0	1
a ₁	0	0	1	1
			b ₁	

$$z_2 = a_1 \bar{a}_0 b_1 + a_1 b_1 \bar{b}_0$$

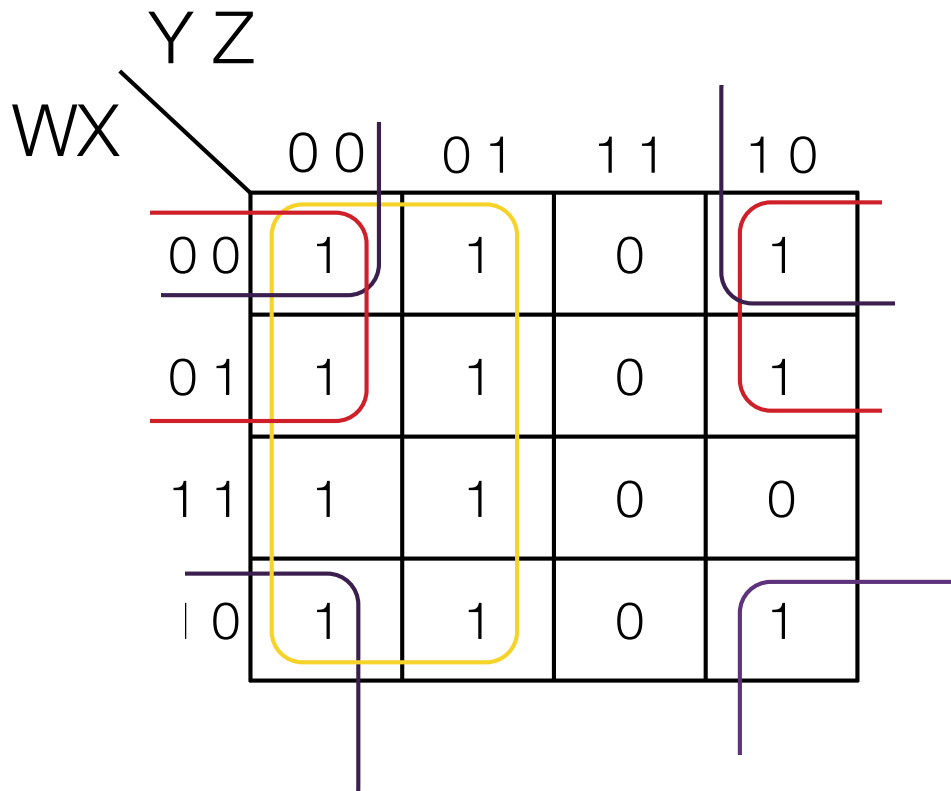
$$z_0 = a_0 b_0$$

			b ₀	
	0	0	0	0
	0	1	1	0
	0	1	1	0
a ₁	0	0	0	0
			b ₁	

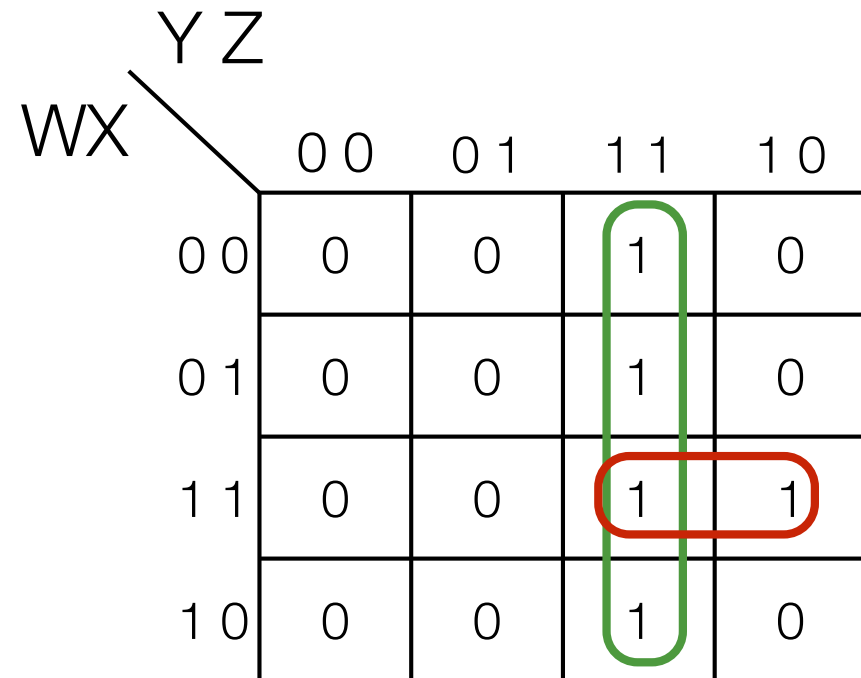
K-Maps: Complements, PoS, don't care conditions

Finding $\bar{F}$

- Find prime implicants corresponding to the 0s on a k-map



$$F = \bar{Y} + \bar{X}\bar{Z} + \bar{W}\bar{Z}$$



$$\bar{F} = YZ + WXY$$

PoS expressions from a k-map

- Find $\bar{F}$ as SoP and then apply DeMorgan's
- or: Cover the boxes with all "0"s - the sum term "0's out" that box

		Y Z			
		0 0	0 1	1 1	1 0
W X	0 0	1	1	0	1
	0 1	1	0	0	0
	1 1	1	0	0	0
	1 0	1	1	0	1

$$\bar{F} = YZ + XZ + YX$$

DeMorgan's

$$F = (\bar{Y} + \bar{Z})(\bar{Z} + \bar{X})(\bar{Y} + \bar{X})$$

Don't Care Conditions

Don't care conditions

- There are circumstances in which the value of an output doesn't matter
- For example, in that 2-bit multiplier, what we
- are told neither a nor b will be input as 0
 - “don't care” what the output looks like for the input cases that will not occur
- Don't care situations are denoted by an “X” in a truth table and in Karnaugh maps.
- Can also be expressed in minterm form:

$$z2 = \sum m(10, 11, 14)$$

$$d2 = \sum m(0, 1, 2, 3, 4, 8, 12)$$

a1	a0	b1	b0	z3	z2	z1	z0
0	0	0	0	X	X	X	X
0	0	0	1	X	X	X	X
0	0	1	0	X	X	X	X
0	0	1	1	X	X	X	X
0	1	0	0	X	X	X	X
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	X	X	X	X
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	X	X	X	X
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

Simple Don't Care Example

- Let $F = AB + \bar{A}\bar{B}$
- Suppose we know the input combo $A=1, B=0$ will never occur
- Can we replace F with a simpler function G whose output matches for all inputs we do care about?
- Let H be the function with Don't-care conditions for obsolete inputs

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

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A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

Inputs will
not occur

Since $A=1, B=0$ will not occur,
don't care what circuit
produces for $A=1, B=0$ inputs

Simple Don't Care Example

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A	B	F	H
0	0	1	1
0	1	0	0
1	0	0	X
1	1	1	1

Inputs will
not occur



Simple Don't Care Example

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A	B	F	H	G
0	0	1	1	1
0	1	0	0	0
1	0	0	X	1
1	1	1	1	1

Inputs will
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Simple Don't Care Example

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A	B	F	H	G
0	0	1	1	1
0	1	0	0	0
1	0	0	X	1
1	1	1	1	1

Inputs will
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- Both F & G are appropriate functions for H

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A	B	F	H	G
0	0	1	1	1
0	1	0	0	0
1	0	0	X	1
1	1	1	1	1

Inputs will
not occur



- Both F & G are appropriate functions for H

$$G = AB + \bar{B}$$

- G is (slightly) simpler than F (3 instead of 4 literals), and gets the job done!

2-bit multiplier non-0 multiplier (SOLUTION)

a1	a0	b1	b0	z3	z2	z1	z0
0	0	0	0	X	X	X	X
0	0	0	1	X	X	X	X
0	0	1	0	X	X	X	X
0	0	1	1	X	X	X	X
0	1	0	0	X	X	X	X
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	X	X	X	X
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	X	X	X	X
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

			b ₀	
	X	X	X	X
	X	0	0	0
a ₁	X	0	1	0
	X	0	0	0
			b ₁	a ₀

$$z_3 = a_1 a_0 b_1 b_0$$

			b ₀	
	X	X	X	X
	X	0	0	0
a ₁	X	0	0	1
	X	0	1	1
			b ₁	a ₀

$$z_2 = a_1 \bar{b}_0 + \bar{a}_0 b_1$$

$$(\text{vs. } a_1 \bar{a}_0 b_1 + a_1 b_1 \bar{b}_0)$$

Revised rule for “implicant”:

1's must be covered

0's must not be covered

X's are optionally covered

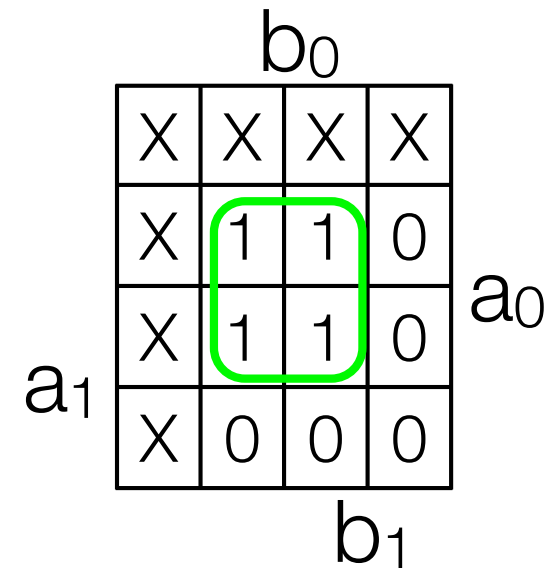
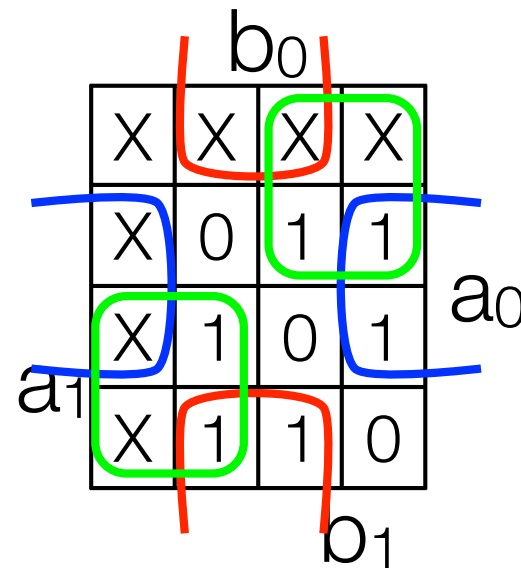
2-bit multiplier non-0 multiplier (SOLUTION)

a1	a0	b1	b0	z3	z2	z1	z0
0	0	0	0	X	X	X	X
0	0	0	1	X	X	X	X
0	0	1	0	X	X	X	X
0	0	1	1	X	X	X	X
0	1	0	0	X	X	X	X
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	X	X	X	X
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	X	X	X	X
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

Still have prime and essential prime implicants

$z_1 = (\text{exercise})$

$z_0 = a_0b_0$



All above prime implicants are essential

Final thoughts on Don't care conditions

- Sometimes “don't cares” greatly simplify circuitry

		D					
		1	X	X	X		
		X	1	X	X	B	
A	0	0	1	X			
	0	0	X	1			
	C						

$$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + ABCD + A\bar{B}C\bar{D} \text{ vs. } \bar{A} + C$$

High-level Summary

- **minterm** (of k variables): a product term formed from the literals of each variable (exist without function)
- A **function** (e.g., for $F = A + \overline{A}BC$) can be constructed by ORing (summing) together the minterms for which the function = 1
- **implicant (of a function)**: a product term (of literals) that is “contained” within the function (formed from minterms where the function = 1)
- **prime implicant**: a product term which, if any literal is remove, is no longer “contained” within the function
- **essential prime implicant**: when a function is expressed as an OR (sum) of prime implicants, this one must be included.

$$AB\overline{C}$$

$$ABC + AB\overline{C} + A\overline{B}C + A\overline{B}\overline{C} + \overline{A}BC$$

$$AB$$